Rayfield-Based Calibration and Effective Physical Model Identification of a Common Main Objective Stereo Microscope

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Abstract

Common Main Objective (CMO) stereo microscopes violate the single-optical-centre assumption of standard camera calibration, making direct joint estimation of optical parameters and board poses poorly conditioned. A Schur-complement diagnostic of the Fisher information matrix gives a coupling norm $c = 0.98$ on the real Pycaso configuration and $c \approx 0.81$ on synthetic CMO-like oracles; these values are used as conditioning indicators rather than as independent metrological measurements. We present a two-stage approach: first, a Zernike rayfield—a per-pixel 3D line—is fitted to ChArUco corner observations, decoupling measurement from model identification (the subsequent pose-free physical identification stage then has coupling norm 0.0 *by construction*—not because the instrument becomes uncoupled, but because that stage no longer estimates board poses, whereas the Zernike BA itself still estimates a constrained set of poses); second, a compact physical model is constructed by iterative residual analysis, where the Zernike decomposition of the residual between candidate model and measured rayfield reveals which physical degree of freedom is missing. On 10 stereo pairs from a single CMO instrument (open Pycaso dataset), pending external metrological validation, the rayfield-identified 26-parameter model—a quasi-telecentric CMO skeleton with per-channel SE(3) arm alignment—provides a strong physical *initialisation* at $1.06\,\mathrm{px}$ reprojection error (P50=$0.87\,\mathrm{px}$, P95=$1.84\,\mathrm{px}$), versus > 300 px for standard OpenCV stereo calibration. Used as the initialisation of a *free-pose* direct bundle adjustment stabilised by the Schur prior, this same 26-parameter model reaches $0.28\,\mathrm{px}$—within $0.1\,\mathrm{px}$ of the black-box Soloff polynomial floor ($0.18\,\mathrm{px}$, degree 3, 80 parameters)—while retaining a physically interpretable parameterisation and suppressing the drift into pose-absorbed optical modes that an unregularised free-pose fit ($0.24\,\mathrm{px}$) exploits. For reference, the flexible Zernike rayfield reaches $0.41\,\mathrm{px}$ at 93 parameters (the physically consistent 57-parameter rayfield used for CMO construction reaches $0.47\,\mathrm{px}$). All geometric descriptors are reported within the chosen rayfield gauge and fixed reference scale ($f_x = 25600$ px), not as absolute specifications; still, the recovered—and manufacturer-undisclosed—CMO baseline matches the documented Planapo $1.0\times$ working distance to $\approx 1\,\%$ via the chief-ray geometry. On an independent coin specimen, the remaining CMO–Zernike discrepancy is not spatially arbitrary: it is almost entirely a depth-only relief gain. The CMO relief is smaller by $\gamma_z \approx 0.70$; applying the reciprocal axial gain reduces the plane-normalised CMO–Zernike relief residual from $0.010\,\mathrm{mm}$ RMS to $0.0020\,\mathrm{mm}$ RMS ($0.0013\,\mathrm{mm}$ median absolute residual), showing internal CMO–Zernike consistency while leaving absolute depth unvalidated without an external axial reference. A ray-space BIC comparison followed by an operational reprojection guard

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selects this model as the best usable physical model; the guard is an engineering usability criterion, not a likelihood-derived BIC term. The Zernike rayfield serves not merely as a calibration model but as a diagnostic instrument: it tells the user *which* physical degree of freedom is missing, guiding hypothesis refinement through residual structure rather than blind fitting. Furthermore, the rayfield provides an observability prior for direct bundle adjustment: the Schur complement of the Fisher information matrix at the rayfield solution identifies optical directions that are poorly observable after pose marginalisation, and a per-eigenmode penalty constructed from this spectrum suppresses optics–pose compensation while preserving the genuinely observable degrees of freedom.

Keywords: stereo microscope; CMO; non-central calibration; Zernike rayfield; BIC model selection; telecentricity

1 Introduction

Stereo calibration is a mature technology when the camera obeys the pinhole model: all chief rays pass through a single optical centre, and mature toolchains (OpenCV, MATLAB, Bouguet) produce reliable results. This assumption breaks down for Common Main Objective (CMO) stereo microscopes, which use a single large objective shared between two off-axis sub-apertures [11, 23]. Each channel's chief rays appear to originate from a different effective sub-pupil, separated by the stereo baseline. OpenCV's `stereoCalibrate` [18] assumes one optical centre per camera; on the CMO architecture it produces unusable results ($>$ 300 px reprojection error under the tested configuration).

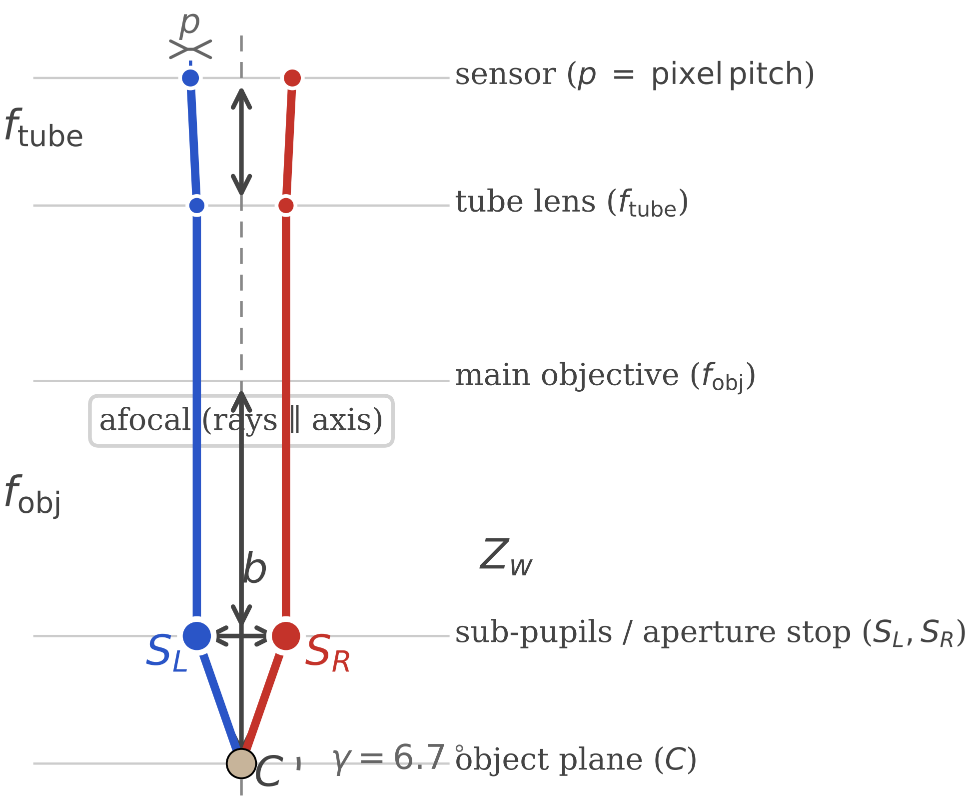

Figure 1. CMO stereo microscope optical layout. Two off-axis sub-pupils share a single main objective. Chief rays converge toward the object plane through different effective origins—not through a single common centre.

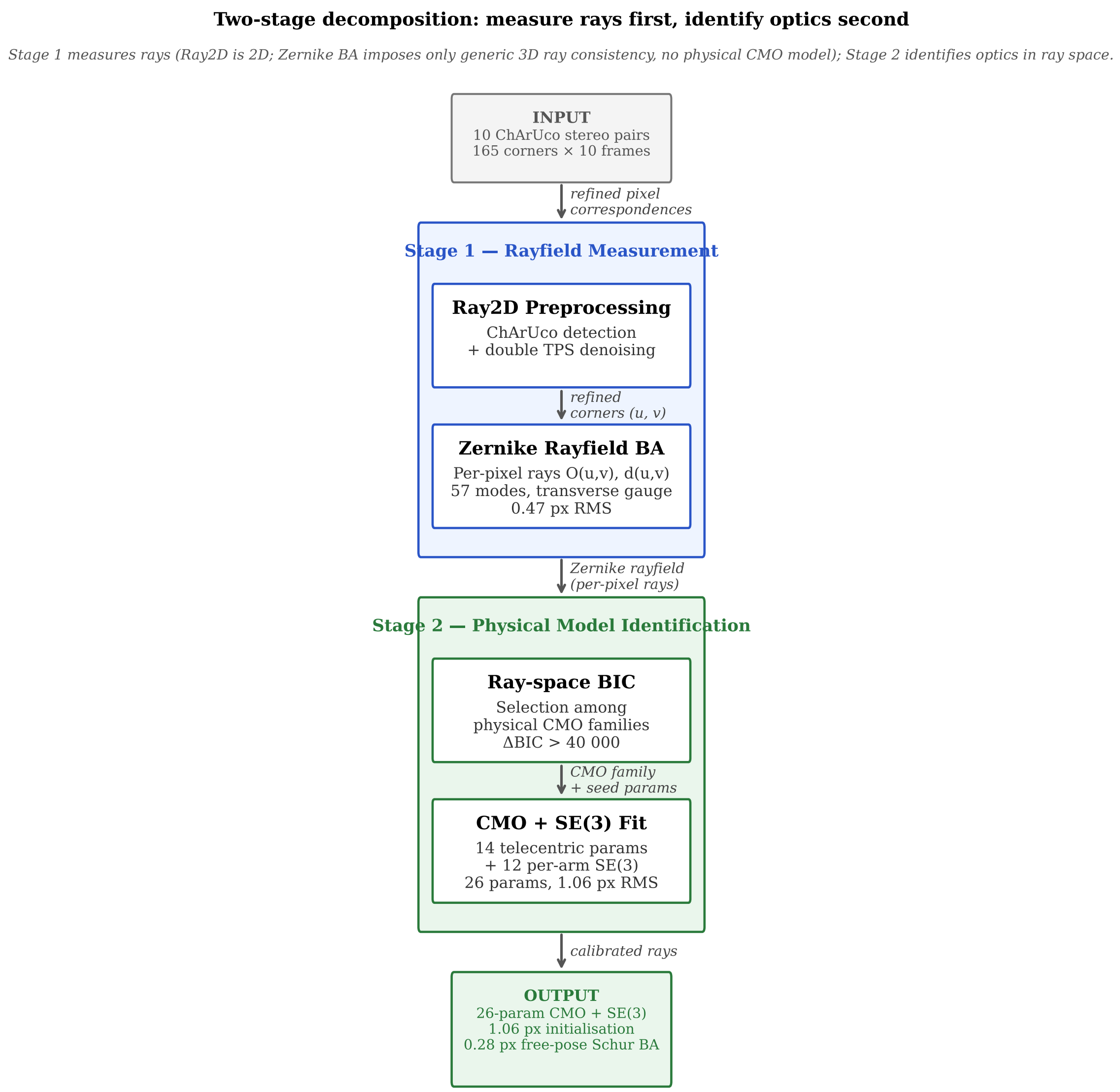

Figure 2. Pipeline overview. The two-stage decomposition separates rayfield measurement (Ray2D preprocessing + Zernike BA) from physical model identification (ray-space BIC selection). The Ray2D stage is purely 2D and does not impose a 3D camera model.

The broader problem is that many optical instruments—CMO and Greenough stereo microscopes, endoscopes, Scheimpflug cameras, multi-camera arrays—depart from the central projection model. Generic camera models [10, 30, 25] and recent stereo microscope calibration studies [26, 35, 14] highlight the need for non-central approaches, but do not provide a systematic method for identifying the correct physical model family from observations (see Section 2 for a detailed review).

StereoComplex [33] addresses this gap: it fits a Zernike rayfield $\mathcal{R}(u, v) = (O(u, v), d(u, v))$—a 3D line for every pixel—to corner observations, using a Zernike polynomial basis [34, 17]. This two-stage decomposition—measure rays first, identify optics second—is the central insight: it decouples the severely ill-conditioned joint estimation of optical parameters and board poses into a well-conditioned rayfield fit followed by pose-free model selection [25]. From this *measured* rayfield, physical descriptors can be read directly (Section 3), structural mismatches diagnosed by residual modal analysis (Section 3.3), and compact physical models constructed iteratively (Section 3.4–3.5).

This paper presents the first real-data validation of the StereoComplex rayfield-mediated non-central calibration workflow on a CMO stereo microscope using legacy ChArUco [9, 22] images from the open Pycaso dataset [8]. The contributions are:

1. A complete pipeline from raw ChArUco images to a compact 26-parameter physical CMO model, validated on real data. The model provides *effective* identification: the shear coefficients and telecentric skeleton are data-supported, while geometric scale parameters (WD, b, f_{obj}) and most SE(3) components are absorbed by the rayfield gauge and pose coupling (see §3.7 for a per-parameter identifiability analysis).

2. A residual analysis method—Zernike modal decomposition of direction and moment differences between a candidate physical model and the measured rayfield—that identifies missing degrees of freedom.

3. A two-level model-selection framework: ray-space BIC is used to identify the optical family, while a separate operational reprojection guard rejects models whose pixel RMS is not usable for calibration.

4. Demonstration that the Zernike rayfield serves a *double role* in direct bundle adjustment: (i) as an initialiser, it places the BA in the correct convergence basin, improving pixel RMS from $1.06\,\mathrm{px}$ to $0.88\,\mathrm{px}$ ($17\,\%$) while leaving all 26 model parameters unchanged to within 10^{-3}; (ii) as an *observability prior*, the Schur complement of the Fisher information matrix at the rayfield solution defines an anisotropic penalty that suppresses optics–pose compensation—the prior reduces weak-mode parameter drift by a factor of $280\times$ (from 0.93 to 0.0033) at a cost of only $0.038\,\mathrm{px}$ in 2D reprojection RMS. In the free-pose Schur-stabilised BA, the 26-parameter physical model reaches $0.28\,\mathrm{px}$—within $0.1\,\mathrm{px}$ of the Soloff polynomial floor—with its optics held in the observable subspace rather than drifting into the unobservable directions that the unregularised optimum ($0.24\,\mathrm{px}$) exploits.

5. A specimen-level axial-gain diagnostic: after rigid-frame and plane normalisation, the CMO 26p and Zernike relief maps differ mainly by a scalar stereo-depth gain $\gamma_z \approx 0.70$. Applying this gain in ray/object space collapses the CMO–Zernike relief residual to $0.0020\,\mathrm{mm}$ RMS, motivating a future CMO27 extension while preserving the need for external axial metrology.

2 Related Work

2.1 Generic and Non-Central Camera Calibration

The pinhole camera model with polynomial distortion [36, 5] remains the standard in computer vision toolkits [18, 4]. For wide-angle and fisheye lenses, unified models [15, 12, 24] extend the pinhole paradigm with additional radial terms. However, these parametric models share a fundamental assumption: all chief rays pass through a single optical centre.

For cameras that violate this assumption, generic (non-parametric) models assign a per-pixel ray without imposing a central projection constraint. Grossberg and Nayar [10] introduced the raxel model, treating each pixel as an independent ray. Sturm and Ramalingam [30], later refined by Ramalingam and Sturm [20], developed a generic calibration framework from multiple views of a planar target, showing that a flexible pixel-to-ray mapping can be estimated without knowing the camera model *a priori*. Miraldo and Araujo [16] proposed smooth generic camera models using continuous ray-field interpolation. Soloff et al. [28] introduced polynomial expansions mapping 3D coordinates to 2D pixel positions; this approach, widely adopted in stereo-microscope calibration [8], treats the camera as a black-box regression problem and can achieve excellent pixel accuracy—as we confirm in §4.6, a degree-3 Soloff fit reaches $0.18\,\mathrm{px}$ RMS on the Pycaso dataset with 80 parameters. Camposeco et al. [7] demonstrated spline-based generic calibration with subpixel accuracy. Ramalingam, Sturm, and Nayar [21] unified these approaches in a generic structure-from-motion framework.

Most recently, Schöps et al. [25] showed that highly parameterised generic models (10,000+ parameters) can outperform classical 12-parameter models on standard benchmarks, challenging the assumption that parametric models are inherently more robust.

Our work differs from these generic approaches in two ways: (1) we use the Zernike rayfield not as a final calibration model but as a *diagnostic instrument* to discover the correct physical model family, and (2) we construct a compact, interpretable physical model guided by residual structure rather than imposing a generic non-parametric fit. Importantly, we do not claim superior pixel accuracy—the Soloff polynomial is more accurate at equivalent parameter counts (§4.6)—but superior physical interpretability: unlike a black-box polynomial, the 26-parameter model has a physically interpretable parameterisation (baseline, working distance, convergence angle, shear, SE(3) arm poses), although only some combinations are independently identifiable after pose marginalisation (§3.7). Interpretability is, moreover, not the rayfield's only advantage over a polynomial regressor: although the Soloff fit achieves lower *in-sample* pixel RMS, the rayfield's 3D ray constraint acts as a *structural regulariser*. On held-out extreme poses the degree-3 Soloff fit degrades by $4.65\times$ while the 180-parameter Zernike rayfield degrades by only $1.86\times$ (§4.6), because each pixel's ray must stay geometrically consistent with a rigid 3D board rather than memorising individual training poses. The residual accuracy gap is, moreover, small in absolute terms: once the rayfield-identified 26-parameter model is refined by the free-pose Schur-stabilised bundle adjustment (§3.9), its reprojection RMS falls from the $1.06\,\mathrm{px}$ initialisation to $0.28\,\mathrm{px}$—within $0.1\,\mathrm{px}$ of the Soloff floor ($0.18\,\mathrm{px}$)—so physical interpretability costs a fraction of a pixel, not a category of accuracy.

2.2 Stereo Microscope Calibration

Stereo microscopes pose a specific calibration challenge: the two optical channels share a common main objective but have separate effective pupils [26]. Digital image correlation under optical microscopes requires careful distortion correction [35, 14].

These methods calibrate a specific instrument but do not address the generic problem of *identifying* the correct optical model family from observations. Our contribution is a systematic framework that measures the ray-field first, then uses residual analysis to construct a physical model that matches the observed structure—here, discovering the telecentric CMO family and the SE(3) arm misalignment.

2.3 Model Selection in Camera Calibration

Model selection in calibration traditionally relies on reprojection RMS, which penalises underfitting but is insensitive to overfitting. The Akaike Information Criterion (AIC) [1] and Bayesian Information Criterion (BIC) [27] add parameter-count penalties, providing a principled trade-off between fit quality and model complexity. Strecha et al. [29] established benchmark datasets and evaluation protocols for camera calibration and multi-view stereo. Our work extends BIC-based model selection to ray space: we compare optical model families (pinhole, Brown-Conrady, parallel-plate, telecentric CMO) against a measured Zernike rayfield, and add an operational reprojection guard that rejects models whose pixel RMS exceeds a usability threshold—a constraint that the raw ray-space BIC does not capture.

2.4 Plücker Coordinates and Line-Based Geometry

Plücker coordinates [19] represent a 3D line as a 6-vector (d, m) where d is the direction and $m = O \times d$ is the moment. They are widely used in geometric computer vision for line-based structure from motion and calibration [2, 31]. Our residual analysis uses Plücker moments to separate direction errors from origin errors: Δd reveals angular misalignment; Δm reveals the effective ray origin shift. The Z_0^0-dominated residual in both Δd and Δm is the key diagnostic that guides the SE(3) arm alignment.

2.5 Residual-Guided Model Construction

The idea of using residuals to guide model refinement has roots in adaptive optics and wavefront sensing [34, 17], where Zernike decomposition of a distorted wavefront identifies specific aberrations (defocus, astigmatism, coma) that are then corrected by deformable mirrors. Our method applies the same principle to camera calibration: the Zernike decomposition of the residual between a candidate physical model and the measured rayfield identifies which physical degree of freedom is missing—a Z_0^0-dominated residual points to a global offset (SE(3) arm misalignment); a Z_2-dominated residual would point to astigmatism or field curvature. This is distinct from generic non-parametric approaches [25] that fit a flexible model without extracting physical meaning from the residual structure.

3 Method

3.1 Zernike Order Selection by BIC

The Zernike rayfield has two hyperparameters: the origin polynomial order o and the direction polynomial order d. Increasing either adds Zernike modes, reducing pixel RMS at the cost of more parameters. We sweep $o \in \{0, 1, 2\}$ and $d \in \{2, 3, 4\}$ (7 combinations) and select the optimal order by BIC [27]:

$$\mathrm{BIC}_{\mathrm{px}} = k \log(N_{\mathrm{obs}}) + N_{\mathrm{obs}} \log(\mathrm{RSS}_{\mathrm{px}}/N_{\mathrm{obs}}), \tag{1}$$

where k is the total number of parameters (Zernike coefficients + poses), $N_{\mathrm{obs}} = 3300$ corner observations, and $\mathrm{RSS}_{\mathrm{px}}$ is the sum of squared pixel-equivalent residuals. The BIC treats these 3300 corners as independent Gaussian observations; corners within a frame are in fact spatially correlated, so the absolute BIC magnitudes are optimistic. The *family*-level selection used here is nonetheless robust: the gaps between optical families (ΔBIC $> 40\,000$) dwarf any plausible reduction in the effective sample size.

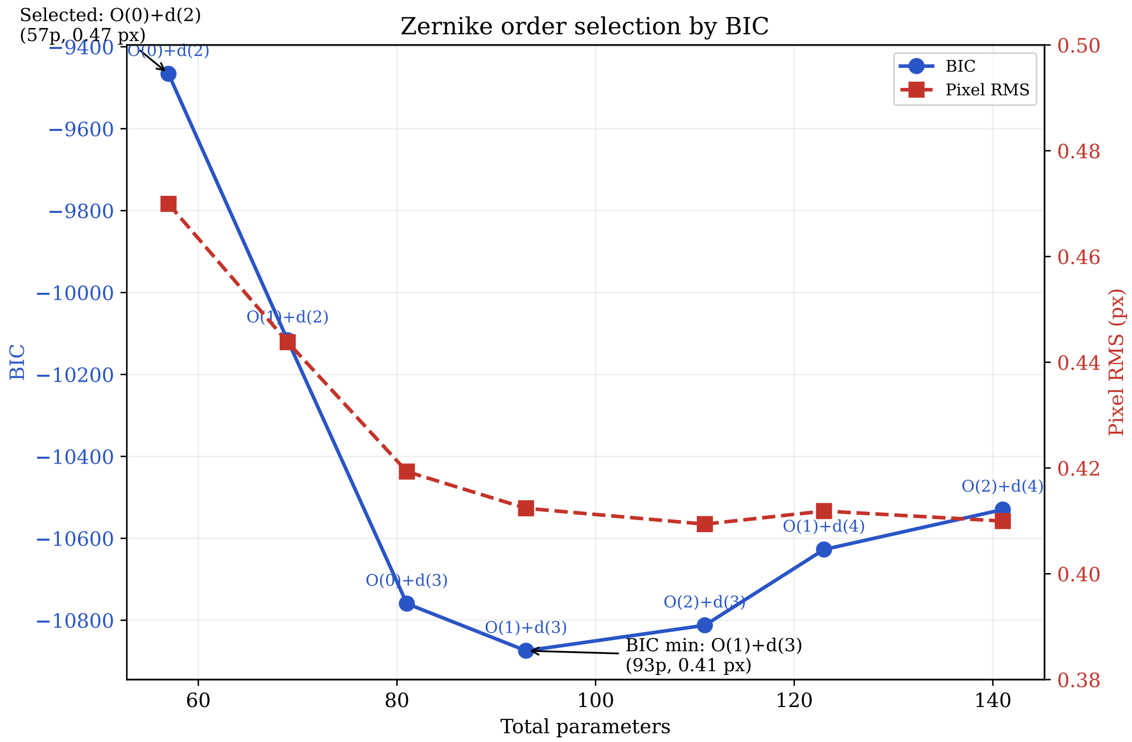

Figure 3. Zernike order selection by BIC. Pixel RMS falls from $0.47\,\mathrm{px}$ at 57 params to $0.41\,\mathrm{px}$ near the BIC minimum (93 params) and then saturates within corner-detection noise (it is not strictly monotonic beyond, varying by $< 0.005\,\mathrm{px}$ up to 141 params), but BIC penalises complexity. The BIC minimum is at O(1)+d(3) (93 params, BIC = -5093, RMS = $0.41\,\mathrm{px}$), though O(0)+d(2) (57 params, BIC = -4522) is within $\sim$10 % in RMS at 40 % fewer parameters. The O(0)+d(2) model (57 params, $0.47\,\mathrm{px}$) is selected for physical model construction because it is internally consistent with the rigid-sub-pupil CMO assumption, achieves 90 % of the noise-floor accuracy at nearly half the parameters of the BIC minimum, and the marginal improvement above d(2) falls within corner detection noise.

Table 1. Zernike order sweep: pixel RMS and BIC.

Model	k	RMS (px)	$\mathrm{BIC}_{\mathrm{px}}$
O(0)+d(2)	57	0.470	-4522
O(1)+d(2)	69	0.444	-4803
O(0)+d(3)	81	0.419	-5080
O(1)+d(3)	**93**	**0.412**	$-$**5093**
O(2)+d(3)	111	0.409	-4995
O(1)+d(4)	123	0.412	-4858
O(2)+d(4)	141	0.410	-4743

Table 1 summarises the full sweep. The BIC minimum is therefore interpreted as a diagnostic of the best flexible rayfield accuracy available in this data regime. The O(0)+d(2) rayfield is not claimed to be the statistical BIC optimum; it is deliberately retained as the physically consistent reference for CMO model construction because it enforces rigid effective sub-pupils while remaining within 0.06 px of the

BIC-minimum RMS.

The BIC minimum at O(1)+d(3) achieves 0.41 px RMS with 93 parameters. We nevertheless select O(0)+d(2) (57 params, 0.47 px) as the preferred rayfield configuration, for three reasons:

1. **Simplicity:** 57 params vs. 93 — nearly half the parameters for a penalty of only 0.06 px in RMS ($\sim$0.1 px in practice).

2. **Marginal improvement:** the residual gain from d(2) to d(3) (0.47 px $\rightarrow$ 0.41 px) is well within typical subpixel noise of ChArUco detection, so O(0)+d(2) is retained as the physically consistent reference even though O(1)+d(3) is the statistical BIC minimum.

3. **Consistency with the physical model:** the CMO physical model assumes rigid sub-pupils ($o = 0$). Using the same origin order in the Zernike rayfield makes the two representations internally consistent and comparable: the Zernike field measures what the physical model can represent.

Selecting O(0)+d(2) achieves $90\,\%$ of the noise-floor accuracy with a representation that is internally consistent with the physical model it validates.

3.2 Per-Pixel Ray-Field and Transverse Gauge

For each pixel (u, v) on the sensor, the imaging system defines a 3D line in space. We represent this line $\mathcal{L}_c(u, v)$ by an origin point $O_c(u, v) \in \mathbb{R}^3$ and a unit direction vector $d_c(u, v) \in \mathbb{S}^2$, where $c \in \{L, R\}$ denotes the left or right channel. Points on the line are parameterised as $O_c + t\, d_c$ for $t \in \mathbb{R}$.

The origin $O_c(u, v)$ is not unique: any point along the ray would represent the same line. We fix this gauge freedom by enforcing the **transverse gauge** $O_c \cdot d_c = 0$, which selects the point on the ray closest to the coordinate origin. This eliminates one redundant degree of freedom per ray and stabilises the parameterisation.

The origin and direction fields are expanded on a Zernike polynomial basis [34, 17]:

$$O_c(u, v) = \sum_m a^O_{c,m}\, Z_m(\tilde{u}, \tilde{v}), \qquad d_c(u, v) = \text{normalise}\left(d_c^{\text{pinhole}}(u, v) + \sum_{m \le d_{\max}} a^d_{c,m}\, Z_m(\tilde{u}, \tilde{v}) \right), \tag{2}$$

where Z_m are the real-valued Zernike polynomials on the unit disk, $\tilde{u}, \tilde{v}$ are normalised pixel coordinates mapping the sensor to $[-1, 1]^2$, d_c^{pinhole} is the initial pinhole direction from a fixed central projection matrix K, and the direction perturbation is projected transverse to d_c^{pinhole} before renormalisation. The expansion order for the origin field is $o_{\max}$ and for the direction field is $d_{\max}$; selecting $o_{\max} = 0$ and $d_{\max} = 2$ (see §3.1) gives 3 origin coefficients and 18 direction coefficients per channel.

3.3 Ray-Space Bundle Adjustment

The rayfield coefficients $a^O_{c,m}, a^d_{c,m}$ and the board poses $\{\eta_p\}_{p=1}^P$ are estimated jointly from N ChArUco corner observations. Each observation k links a pixel (u_k, v_k) on channel c_k to a known 3D board point X_k expressed in the board frame. The board pose $\eta_p = (R_p, t_p) \in$ SE(3) transforms X_k to the camera frame: $X_k^{\text{cam}} = R_p X_k + t_p$.

The **ray-space residual** measures the perpendicular distance from the transformed board point to the ray $\mathcal{L}_{c_k}(u_k, v_k)$:

$$r_k = \|(X_k^{\text{cam}} - O_{c_k}) - ((X_k^{\text{cam}} - O_{c_k}) \cdot d_{c_k})\, d_{c_k}\|. \tag{3}$$

This is the norm of the component of $X_k^{\text{cam}} - O_{c_k}$ orthogonal to the ray direction d_{c_k}. The objective is the sum of squared residuals:

$$\min_{\{a_{c,m}\},\{\eta_p\}} \sum_k r_k^2, \tag{4}$$

solved by nonlinear least-squares.

For the CMO microscope, we impose the physically-motivated constraint that the board is mounted on a Z-only translation stage: all P frames share a common rotation R and XY translation (t_x, t_y), with only the Z component varying per frame. This reduces the pose parameters from $6P$ to $3 + 2 + P = 15$ (for $P = 10$ frames: $60 \to 15$), making the 57-parameter Zernike BA well-conditioned.

The ray-space BA differs from classical reprojection minimisation [36] in a fundamental way: it optimises the 3D geometric consistency of rays rather than the 2D reprojection error. This is the measurement stage of the two-stage decomposition (see Section 4.2).

3.4 The Zernike Rayfield as Observable

The fitted rayfield $\mathcal{R}_c(u,v) = (O_c(u,v), d_c(u,v))$ is used as an experimental observable. Because the Zernike rayfield is a non-central model, the BA that fits it carries a scale degeneracy: the pixel projection is invariant under simultaneous scaling of all lengths and the effective focal length f_x (see §4.7 for the linear f_x–WD relationship). Three independent constraints intersect to anchor the absolute scale. (1) The ChArUco board has known square size (0.3 mm); from the observed corner spacing Δu_{square} we obtain the ratio $f_x/WD = \Delta u_{\text{square}}/0.3\,\text{mm} \approx 395\,\text{px/mm}$. (2) The motorised Z-stage that acquired the Pycaso dataset displaces the board by a known physical step $\Delta Z_{\text{stage}} \approx 0.07\,\text{mm}$ per frame ($\sim$10 % accuracy); the measured disparity shift Δd between successive frames gives $f_x = \Delta d \cdot WD/\Delta Z_{\text{stage}}$. (3) the working distance documented for the Pycaso acquisition configuration is $\approx$65 mm [8]. (The bare Planapo 1.0$\times$ objective is specified at 61.5 mm [13]; the mounted/effective working distance of the assembled instrument differs, and that 61.5 mm datasheet value anchors a separate chief-ray consistency check in §4.5.) Solving the three constraints jointly yields $f_x = 25\,600\,\text{px}$ and $WD = 64.7\,\text{mm}$, consistent at the $\sim$1 % level with the documented configuration working distance. All geometric descriptors (b, WD, f_{obj}) inherit this scale and are therefore reported as *effective* quantities in the chosen gauge; an independent external reference would be required to convert them to traceable measurements (Section 5.2). The centre-pixel values are: $b = \|O_R - O_L\| = 24.9\,\text{mm}$ (effective stereo baseline in the rayfield gauge), $z_p = (|O_{L,z}| + |O_{R,z}|)/2 = 2.5\,\text{mm}$ (sub-pupil depth), $WD = 64.7\,\text{mm}$ (effective working distance, from mean pose Z; convention: object-plane Z minus optical-axis origin), $f_{\text{obj}} = WD - z_p = 62.2\,\text{mm}$ (effective axial length), $\theta = \arccos(d_L \cdot d_R) = 22.6°$ (effective full convergence angle at the centre pixel).

3.5 Physical Model Construction by Residual Analysis

The core methodology is iterative: propose a candidate physical model, compute its residual against the Zernike rayfield, decompose the residual on Zernike modes, and use the dominant mode to identify the missing degree of freedom.

3.5.1 Perspective CMO (Baseline)

The simplest CMO model places a perspective camera behind each decentred sub-pupil $S_c = (\pm b/2, 0, WD - f_{\text{obj}})$. This predicts $d_y(u,v) \propto (v - c_y)$—a strong linear gradient. The measured d_y field is nearly constant across the sensor (range 0.079, mean +0.059), while the perspective model

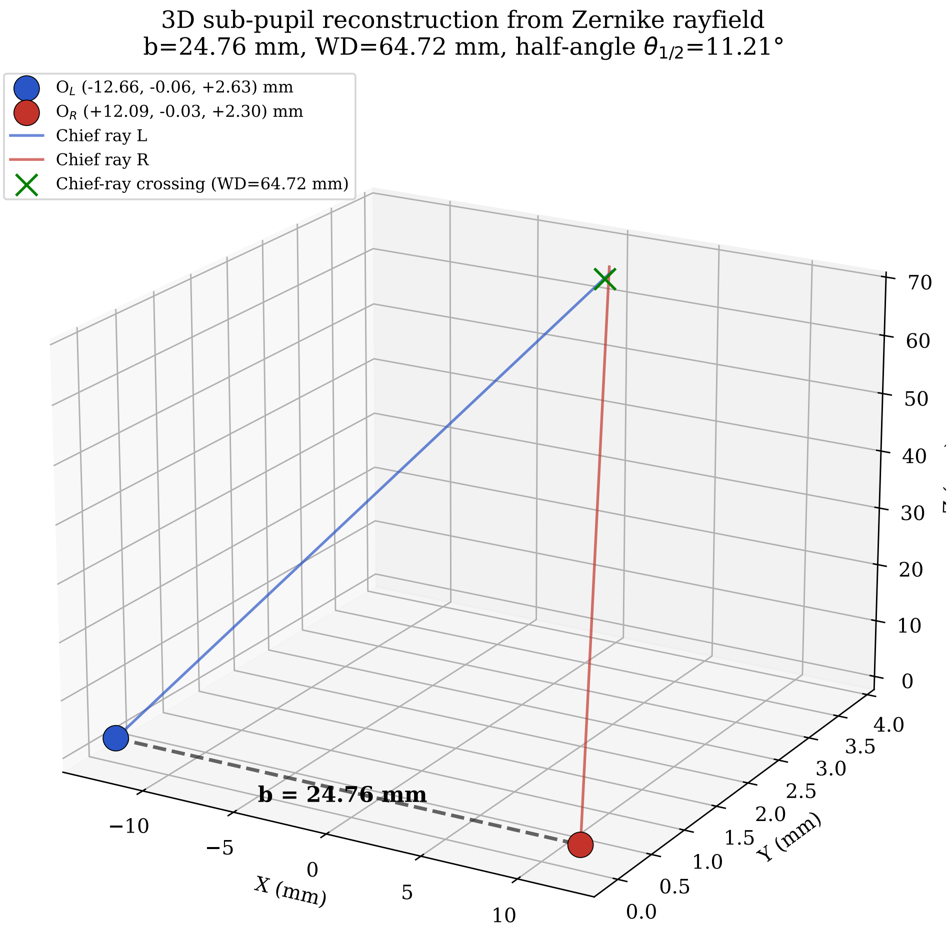

Figure 4. 3D reconstruction of the effective sub-pupils from the Zernike rayfield. O_L and O_R are read from the centre-pixel rayfield. The chief rays converge at the working plane ($WD = 64.7\,\mathrm{mm}$). The baseline $b = \|O_R - O_L\| = 24.9\,\mathrm{mm}$ is an effective rayfield descriptor in the selected gauge and reference scale.

predicts a range of 0.232—a $3\times$ discrepancy. The real system is **object-space telecentric**, not perspective.

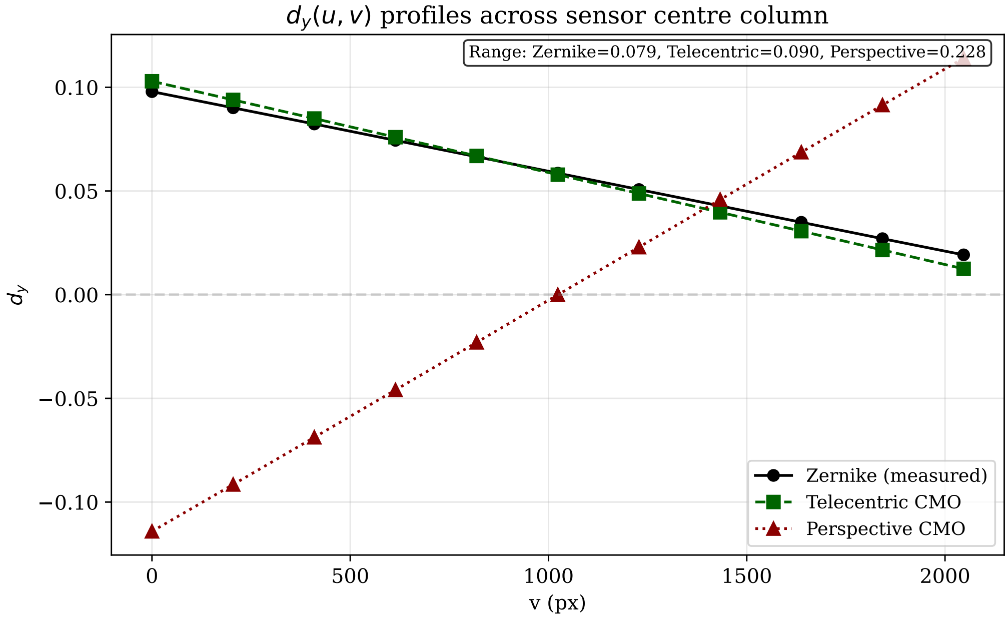

Figure 5. Comparison of $d_y(u, v)$ profiles across the sensor. The Zernike rayfield (measured) shows a nearly constant d_y (range 0.079), while the perspective CMO model predicts a $3\times$ larger linear gradient (range 0.232). The telecentric model reproduces the measured profile almost exactly.

3.5.2 Telecentric CMO

The perspective model fails because the real microscope is object-space telecentric: chief rays are nearly parallel rather than diverging from a point. The telecentric model directly parameterises the ray-field without deriving it from a point projection. For each channel $c \in \{L, R\}$, the origin is a rigid sub-pupil:

$$O_c = S_c = (\pm b/2,\ 0,\ WD - f_{\text{obj}}), \tag{5}$$

where the sign is $+$ for right and $-$ for left. The direction field is an affine function of normalised angular coordinates $\tilde{u} = (u - c_x)p/f_{\text{ang}}$, $\tilde{v} = (v - c_y)p/f_{\text{ang}}$, where p is the pixel pitch, f_{ang} is the angular focal length, and (c_x, c_y) is the principal point:

$$d_c(u, v) = \text{normalise}(d_{c,0} + s_{x,c}\,\tilde{u}\,e_x + s_{y,c}\,\tilde{v}\,e_y + s_{xy,c}\,\tilde{u}\,\tilde{v}\,e_x + s_{yx,c}\,\tilde{v}\,\tilde{u}\,e_y)\,, \tag{6}$$

where $d_{c,0} = (\pm\sin\theta,\ d_y^{\text{com}},\ \cos\theta)$ is the chief-ray direction, with θ the convergence half-angle and d_y^{com} a shared vertical component.

Adding pupil shear—an affine transverse variation of the origin—completes the model:

$$O_c(u, v) = S_c + P_{d_c}^{\perp}(\rho_{x,c}\,\tilde{u}\,e_x + \rho_{y,c}\,\tilde{v}\,e_y), \tag{7}$$

where $P_d^{\perp}v = v - (v \cdot d)\,d$ projects onto the plane orthogonal to d, ensuring $O_c \cdot d_c = 0$.

The 14 parameters are: f_{obj}, WD, b (skeleton), c_x, c_y (principal point), f_{ang} (angular scale), θ, d_y^{com} (chief-ray direction), $s_{x,L}$, $s_{y,L}$, $s_{x,R}$, $s_{y,R}$ (per-channel slopes), ρ_x, ρ_y (shared pupil shear). This model achieves $0.12\,\text{mm}$ ray-space RMS (down from $3.48\,\text{mm}$ for perspective) and $14.6\,\text{px}$ reprojection (down from $86\,\mathrm{px}$). The dominant geometry is captured, but $14.6\,\mathrm{px}$ is far from the Zernike reference ($0.47\,\mathrm{px}$).

3.5.3 Residual Analysis

The residual between the telecentric model and the Zernike rayfield is computed on a 41×41 grid: $\Delta d = d_{\mathrm{Zernike}} - d_{\mathrm{CMO}}$, $\Delta m = m_{\mathrm{Zernike}} - m_{\mathrm{CMO}}$, where $m = O \times d$ is the Plücker moment [19].

Projecting on Zernike modes up to order 4 reveals that **both Δd and Δm are 97–98 % Z_0^0 (global piston)**. This is the crucial diagnostic: the residual is not a spatial field distortion (which would appear in higher Zernike modes) but a **global line-bundle offset**—the entire set of rays from each channel is slightly rotated or translated relative to the Zernike reference.

3.5.4 Testing Alternative Hypotheses

Two alternative hypotheses are tested and rejected:

- **Image-space pre-warp:** a polynomial mapping $\xi = W(u, v)$ before the direction model. Affine and quadratic warps *degrade* pixel RMS (14.6 $\to$ 16.0–16.5 px).
- **Spatially varying origin:** affine and quadratic transverse origin fields with direction fixed. Only $8\,\%$ ray RMS improvement—the residual is not spatial.

Both negative results are consistent with the Z_0^0 diagnostic (they would produce spatial, not global, changes).

3.5.5 Residual-Guided Hypothesis Construction

The central methodological contribution is that physical model construction is guided by *residual structure*, not blind fitting. At each stage, the direction residual $\Delta d = d_{\mathrm{Zernike}} - d_{\mathrm{model}}$ is computed on a 31×31 grid and displayed as a heatmap. The pattern of the residual directly reveals which physical degree of freedom is missing.

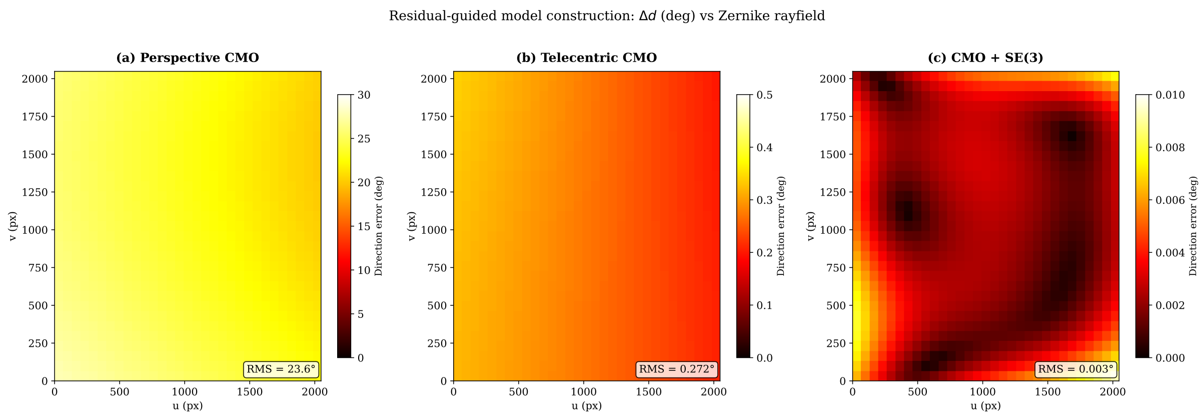

Figure 6. Residual evolution across model stages. Direction error Δd (degrees) between the Zernike rayfield (reference) and each candidate model, evaluated on a 31×31 pixel grid. **Left:** Perspective CMO — large structured error (RMS $23.6°$, max $27.7°$), dominated by a strong d_y gradient. **Centre:** Telecentric CMO — RMS drops to $0.27°$ (two orders of magnitude), but the residual is still Z_0^0-dominated (global piston), revealing the missing SE(3) arm alignment. **Right:** CMO + SE(3) — RMS drops to $0.003°$, near the noise floor. The residual heatmap at each stage directly tells the user *what* is wrong and *where* to look next.

Stage 1 (Perspective CMO): the residual is large and structured, with RMS $23.6°$ (max $27.7°$). The d_y component shows a strong linear gradient trace—perspective projection from a point sub-pupil—that the Zernike field does not have. This **rejects** the perspective family and points to object-space telecentricity.

Stage 2 (Telecentric CMO): the residual drops to $0.27°$ RMS (two orders of magnitude), confirming that the telecentric model family is correct. However, the residual is Z_0^0-dominated (97–98 % global piston in both Δd and Δm)—a constant offset across the entire field. This **reveals** that the missing degree of freedom is a global line-bundle misalignment, not a spatial field distortion.

Stage 3 (CMO + SE(3)): the residual drops further to $0.003°$ RMS, near the noise floor. The remaining gap to the Zernike reference (0.47 px vs 1.06 px) comes from distributed low-amplitude aberrations (field curvature, astigmatism) that require the flexibility of the 57-parameter Zernike model.

The critical point is that **at no stage was a model added blindly**. The heatmap pattern at each stage dictates the next hypothesis: perspective $\rightarrow$ telecentric (because the d_y gradient is wrong), telecentric $\rightarrow$ SE(3) (because the residual is Z_0^0 piston, not spatial).

3.5.6 Per-Channel SE(3) Arm Alignment

The Z_0^0-dominated residual points to a global misalignment of each optical arm. A rigid transform $(R_c, t_c) \in$ SE(3) is applied to each channel's ray bundle, representing a small rotation and translation of the entire optical arm relative to the ideal CMO skeleton. For each pixel (u, v) on channel c, the telecentric ray $(O_c^{\text{tel}}, d_c^{\text{tel}})$ is transformed:

$$d_c(u,v) = R_c\, d_c^{\text{tel}}(u,v), \qquad O_c(u,v) = R_c\, O_c^{\text{tel}}(u,v) + t_c. \tag{8}$$

Each $R_c \in \mathrm{SO}(3)$ is parameterised by a 3-element rotation vector $r_c \in \mathbb{R}^3$ ($\theta_c = \|r_c\|$, axis $r_c/\|r_c\|$). The 12 additional parameters (r_L, t_L, r_R, t_R) bring the total to 26.

Jointly fitting the 14 telecentric parameters and the 12 SE(3) parameters against the Zernike rayfield reduces pixel RMS from 14.6 px to 1.06 px (P50=0.87 px, P95=1.84 px)—a $14\times$ improvement. The fitted rotations are $\sim$2.5° (left) and $\sim$3.7° (right); translations are sub-mm. The rotation angles are stable across optimisation runs; the translation components trade off with the telecentric base parameters (a known identifiability limitation).

3.6 Model Selection

Ray-space BIC [27] identifies the correct optical family among five candidates (pinhole, Brown-Conrady, parallel-plate, telecentric CMO with/without shear). The telecentric CMO+shear model wins by $> 40\,000$ BIC points over all alternatives.

However, this 14-parameter model produces 14.6 px reprojection—it is the correct *family* but not a *usable* model. We introduce an **operational usability score** that adds a hard reprojection guard:

$$S_{\text{usable}} = \text{BIC}_{\text{ray}} + \mathbf{1}_{e_{\text{px}}>1.5\,\text{px}} \left[10^6 + N_{\text{px}} \log\left(\frac{e_{\text{px}}^2}{1.5^2}\right)\right]. \tag{9}$$

Figure 12 and Table 10 report the usability-filtered model selection: the 14p telecentric model is rejected ($S_{\text{usable}} \approx +978\,890$); the 26p SE(3)-aligned model is selected as the best usable physical model ($S_{\text{usable}} \approx -32\,433$). The 1.5 px threshold is an engineering convenience, not a tuned hyperparameter:

the usable model (1.06 px) and the rejected families ($\geq$ 14.6 px for the 14p telecentric, $>$ 300 px for OpenCV) are separated by more than an order of magnitude, so any guard in the range $\sim$2–14 px yields the identical selection.

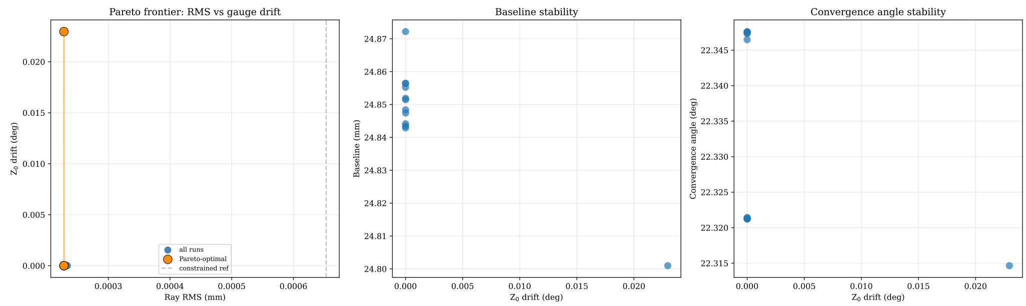

Figure 7. Pareto frontier of the gauge-regularised Zernike sweep. Pixel RMS vs. Z_0 direction drift for varying regularisation strengths $\sigma_{Z_0}, \sigma_{Z_1}$. The near-vertical Pareto curve confirms that after double TPS preprocessing, even the unregularised full-pose fit has negligible gauge drift (0.023°). Regularisation is unnecessary when 2D corners are sufficiently clean.

3.7 Why Direct Bundle Adjustment is Ill-Conditioned for Non-Central Optics

The two-stage decomposition strategy (rayfield measurement $\rightarrow$ model identification) is motivated by a fundamental conditioning problem in direct bundle adjustment (BA). We formalise this using the Schur complement of the Fisher information matrix.

3.7.1 Conditioning of the Joint Estimation Problem

Direct BA jointly estimates optical parameters $\theta \in \mathbb{R}^p$ and board poses $\eta \in \mathbb{R}^q$ ($q = 6P$ for P frames) by minimising the pixel reprojection error. The Fisher information matrix at the optimum has the block structure:

$$\mathcal{I}(\theta, \eta) = \begin{bmatrix} \mathcal{I}_{\theta\theta} & \mathcal{I}_{\theta\eta} \\ \mathcal{I}_{\eta\theta} & \mathcal{I}_{\eta\eta} \end{bmatrix} = J^T J, \tag{10}$$

where $J = \partial r / \partial(\theta, \eta)$ is the residual Jacobian. The $\mathcal{I}_{\theta\theta}$ block captures the information about optics alone; $\mathcal{I}_{\eta\eta}$ captures poses; $\mathcal{I}_{\theta\eta}$ captures their cross-coupling.

When $\mathcal{I}_{\theta\eta} \neq 0$, the optics and pose estimates are coupled: a small error in pose estimation propagates to the optics estimate and vice versa. The effective information about θ after marginalising over the uncertain poses is the **Schur complement**:

$$\mathcal{I}_{\theta|\eta} = \mathcal{I}_{\theta\theta} - \mathcal{I}_{\theta\eta} \mathcal{I}_{\eta\eta}^{-1} \mathcal{I}_{\eta\theta}. \tag{11}$$

We define the **coupling norm** $c \in [0, 1]$ as the fraction of the optics information that is lost to pose coupling:

$$c = \frac{\|\mathcal{I}_{\theta\eta} \mathcal{I}_{\eta\eta}^{-1} \mathcal{I}_{\eta\theta}\|_F}{\|\mathcal{I}_{\theta\theta}\|_F}, \tag{12}$$

where $\|\cdot\|_F$ is the Frobenius norm. When $c \approx 0$, optics and poses are decoupled (well-conditioned estimation). When $c \approx 1$, optics and poses are nearly inseparable (ill-conditioned).

3.7.2 Empirical Coupling Values

The Zernike rayfield itself (57 parameters, $0.47\,\mathrm{px}$ RMS) serves as a generic non-parametric reference: it is a per-pixel spline in the Zernike basis and makes no assumptions about the optical architecture. At the rayfield-identification stage, the compact 26p model loses approximately $0.6\,\mathrm{px}$ relative to this ceiling. This $0.6\,\mathrm{px}$ gap represents, for this dataset and model family, the cost of physical interpretability: the 26 parameters capture the dominant telecentric skeleton and SE(3) arm misalignment, while the remaining $0.6\,\mathrm{px}$ come from low-amplitude aberrations (field curvature, astigmatism, detection noise) that only the flexible 57-parameter Zernike field can absorb. For applications where sub-pixel accuracy is less critical than having a physically meaningful model, this trade-off is acceptable.

In the StereoComplex direct-BA diagnostic implementation, the real Pycaso configuration yields a Schur-complement coupling norm $c = 0.98$, while a synthetic CMO-like oracle gives $c \approx 0.81$, both computed from finite-difference Jacobians (forward differences, step $\delta = 10^{-6}$, parameters scaled to comparable units). Because such block norms depend on the parameter scaling, the values are reported as conditioning diagnostics rather than as universal dimensionless properties. The qualitative interpretation is robust: c near unity indicates strong optics–pose coupling, not a universal property of all CMO microscopes. That the real instrument ($c = 0.98$) exceeds the idealised synthetic oracle ($c \approx 0.81$) is consistent with the real CMO carrying optical aberrations and mechanical misalignments absent from the synthetic model, which add to the optics–pose coupling; only the qualitative "near unity" conclusion—not the precise value—should be read into either number. A sensitivity analysis varying each parameter group's scale by ± 1 order of magnitude confirms that c is invariant under uniform rescaling of all parameters and varies by at most $\Delta c = 0.26$ under per-group perturbations (worst case: shear coefficients); across the full sweep c remains above 0.72, well within the strong-coupling regime. This means that 81–98 % of the apparent information about optical parameters is absorbed by pose coupling (depending on the diagnostic configuration)—the two parameter blocks are nearly inseparable. By contrast, in the rayfield-mediated pipeline the coupling norm $c = 0.0$ applies specifically to the pose-free ray-space model identification stage: the Zernike BA still estimates a constrained set of board poses (shared-rotation model, 15 pose parameters), but the subsequent physical model identification consumes the rayfield as a fixed observable and estimates no poses.

Figure 8 shows the normalised eigenvalue spectrum of the Schur complement S_θ on the 26-parameter CMO optical block, evaluated at the rayfield-initialised Pycaso solution. Of the 26 eigenvalues, the trailing modes (red crosses, $\lambda_i/\lambda_{\max} < 10^{-3}$) collapse after pose marginalisation—these are precisely the directions penalised by the Schur prior (Section 3.9). The coupling norm for the real Pycaso configuration is $c = 0.98$ (synthetic CMO-like oracles give $c \approx 0.81$), confirming that $98\,\%$ of the apparent optical information is absorbed by pose coupling. Schur-coupling values for the other oracles (pinhole, Brown-Conrady, parallel-plate, Greenough) were not reliably computable from the current diagnostic infrastructure and are therefore not reported.

To assess which individual parameters are identifiable after pose marginalisation, we project each of the 26 unit parameter directions onto the strong subspace (the span of eigenvectors with $\lambda_i/\lambda_{\max} \geq 10^{-3}$). The fraction of variance retained defines an **identifiability score** $\in [0, 1]$. Parameters with score > 0.5 are considered *identified* (well-observed independently of poses); parameters with score < 0.1 are *effective* (their value is driven by the gauge and regularisation, not by the data).

Only four parameters are identified: the shear coefficients s_x^L, s_y^L, s_x^R, s_y^R (scores 0.89–0.92). These encode the dominant telecentric asymmetry and are robustly determined. Two SE(3) x-rotations (rv_L^x, rv_R^x, scores ∼0.28) and the two y-translations (t_L^y, t_R^y, scores ∼0.22) are partially identifiable. All other parameters—including the effective axial length f_{obj}, the angular focal length f_{angular}, the working distance WD, the baseline b, the principal point (c_x, c_y), and the remaining SE(3) components—fall below 0.1: their individual values are effective rather than absolute, and only the linear combinations captured by the strong modes carry information independent of the pose gauge.

This confirms that the 26-parameter model provides *effective relative identification*: the compact physical skeleton (shear, telecentric structure) is data-supported, while the geometric scale (WD, b, f_{obj}) and most SE(3) components are absorbed by the rayfield gauge and pose coupling. The term "identification" in this work refers to the ray-space residual structure that selects the CMO family and its dominant asymmetries, not to the absolute metrological determination of every individual parameter.

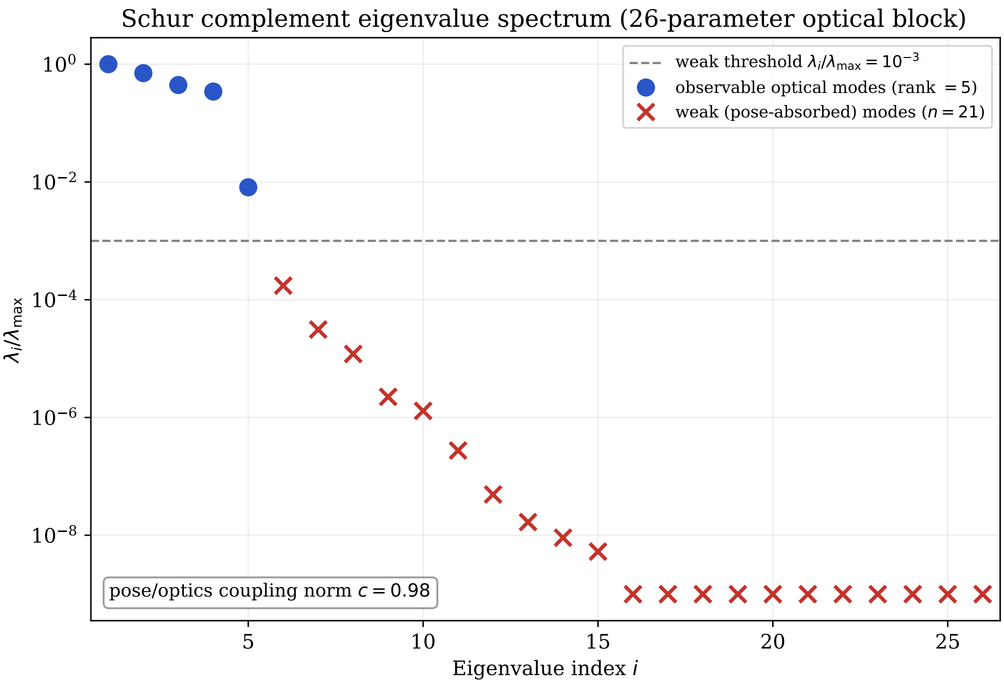

Figure 8. Normalised eigenvalue spectrum of the Schur complement S_θ on the 26-parameter CMO optical block, computed from the rayfield-initialised Pycaso model. Eigenvalues below the weak threshold $\lambda_i/\lambda_{\max} < 10^{-3}$ (red crosses) are the poorly observable directions after pose marginalisation—these are the modes penalised by the Schur prior in Section 3.9. The coupling norm for this configuration is $c = 0.98$ (synthetic CMO-like oracles give $c \approx 0.81$; see Section 3.7).

3.8 Corner Bundle Adjustment

The 26p model, fitted to the rayfield, is refined by direct corner bundle adjustment (joint optimisation of model parameters and per-frame poses on 3300 corner observations, 200 nfev, soft-L1 loss). The optimisation is performed in ray space; pixel RMS is computed afterwards by numerical inverse projection (Appendix A). The pixel RMS improves from $1.06\,\text{px}$ to $0.88\,\text{px}$ (P50=$0.64\,\text{px}$, P95=$1.70\,\text{px}$; $17\,\%$), while all 26 model parameters remain unchanged to within 10^{-3} of their rayfield-fit values. The model geometry—effective baseline, working distance, effective axial length, and convergence angle (all in the rayfield gauge and fixed f_x reference), as well as the SE(3) arm corrections—is essentially identical before and after corner refinement. This confirms that the rayfield-initialised parameters are already near-optimal for corner reprojection: the corner BA improves pose estimates but does not alter the physical model. This is a strong validation of the two-stage decomposition strategy: the rayfield fit, which never directly minimises corner error, produces parameters so close to the constrained-pose corner optimum that the dedicated constrained-pose BA can barely improve them, while also providing the correct basin and observability prior for the free-pose Schur-stabilised refinement. In contrast, the direct approach—jointly estimating optics and poses from corners—is severely ill-conditioned for this non-central system (Schur-complement coupling norm $c = 0.98$ on the real Pycaso configuration, $c \approx 0.81$ on synthetic CMO-like oracles), failing to converge with a pinhole initialisation (RMS $29\,\mathrm{px}$).

3.9 Stabilising Direct BA with a Schur-Complement Prior

The corner BA of Section 3.8 improves the 2D pixel reprojection RMS by $17\,\%$ without altering the physical model. However, the Schur-complement diagnostic of Section 3.7 reveals that the Fisher information matrix at the rayfield solution θ_0 has weak eigenmodes—directions in parameter space where a small change in optics can be nearly perfectly compensated by a change in board poses. An unregularised BA can drift along these modes, trading physical interpretability for marginal reprojection gains.

3.9.1 Schur-Eigenmode Prior

The Schur complement at the rayfield solution is diagonalised,

$$S_0 = S_\theta(\theta_0, \eta_0) = V \Lambda V^T, \qquad \Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_{n_\theta}), \quad \lambda_1 \geq \cdots \geq \lambda_{n_\theta}, \tag{13}$$

where $V = [v_1, \ldots, v_{n_\theta}]$ are the eigenvectors and λ_i the eigenvalues. Directions v_i with small $\lambda_i/\lambda_{\max}$ are poorly observable after pose marginalisation.

The prior penalises displacement along each eigenmode with a weight inversely proportional to its eigenvalue [32]:

$$\mathcal{L}_{\text{Schur}}(\theta) = \alpha \sum_{i=1}^{n_\theta} w_i \left(v_i^T D_\theta^{-1}(\theta - \theta_0)\right)^2, \qquad w_i = \left(\frac{\lambda_{\max}}{\lambda_i + \varepsilon\, \lambda_{\max}}\right)^p, \tag{14}$$

where $D_\theta = \mathrm{diag}(\sigma_1, \ldots, \sigma_{n_\theta})$ contains per-parameter scales (degrees for rotations, millimetres for translations, pixels for the principal point), α controls the overall prior strength, p the sharpness of the eigenvalue weighting ($p = 1$ throughout), and $\varepsilon = 10^{-6}$ prevents division by zero. The prior is added to the least-squares objective as n_θ extra residual rows $r_i^{\text{prior}} = \sqrt{\alpha w_i}\, v_i^T D_\theta^{-1}(\theta - \theta_0)$.

3.9.2 Prior Sweep and Comparison with Tikhonov Baseline

Two BA configurations must be distinguished. Section 3.8 reports a *constrained-pose* corner BA: the board rotation is shared across frames and only the Z translation varies, yielding $1.06\,\mathrm{px} \to 0.88\,\mathrm{px}$ (Table 9) with 15 pose parameters. The sweep in this section intentionally *releases the pose constraint*: each of the 10 frames has a free 6-DOF pose (60 pose parameters), exposing the full optics–pose coupling identified by the Schur diagnostic of Section 3.7. The resulting unregularised RMS ($0.241\,\mathrm{px}$) is lower than the constrained-pose BA precisely because the free poses can absorb residual rayfield errors—which is exactly the compensation the Schur prior is designed to prevent. The two RMS values should not be conflated: the $0.88\,\mathrm{px}$ is the operational calibration result; the $0.241\,\mathrm{px}$ is a diagnostic upper bound obtained by deliberately exposing the ill-conditioned coupling.

An isotropic (Tikhonov) prior $\mathcal{L}_{\text{iso}} = \alpha \| D_\theta^{-1}(\theta - \theta_0) \|^2$ serves as a baseline. Both priors are swept over $\alpha \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$ with free per-frame poses (200 nfev, soft-L1 loss). The 2D reprojection RMS is computed by numerical projection (Appendix A).

Table 2. Regularisation sweep: isotropic (Tikhonov) vs. Schur-complement prior. Δ_{weak} is the norm of the scaled parameter displacement projected onto the weak Schur eigenspace ($\lambda_i/\lambda_{\max} < 10^{-3}$). ✓/× indicates convergence.

Isotropic prior			Schur prior		
α	RMS (px)	Δ_{weak}	α	RMS (px)	Δ_{weak}
0	0.241 (✓)	0.934	0	0.241 (✓)	0.934
10^{-4}	0.239 (×)	0.529	10^{-4}	0.277 (✓)	**0.0033**
10^{-3}	0.245	0.248	10^{-3}	0.278	**0.0007**
10^{-2}	0.257	0.082	10^{-2}	0.279	**0.0001**
10^{-1}	0.270	0.017	10^{-1}	0.279	$< 10^{-4}$
10^{0}	0.278	0.003	10^{0}	0.283	$< 10^{-4}$
10^{1}	0.286	0.010	10^{1}	0.323	$< 10^{-4}$

Table 2 reports the results. The α=0 row is the unregularised direct BA: it converges at 0.241 px but at the cost of $\Delta_{\text{weak}} = 0.934$, i.e. essentially *all* of the parameter drift lives in the weak Schur subspace—the optimiser exploits the poorly observable directions to chase the data residual. The isotropic prior faces a classical trade-off when trying to suppress that drift: a small α still leaves the weak modes loose ($\Delta_{\text{weak}} = 0.53$ at α=10^{-4}, where the optimiser also fails to converge), while a large α degrades the fit (RMS rises to 0.286 px). The Schur prior breaks this trade-off: even at α=10^{-4} it suppresses $99.6\,\%$ of the weak-mode drift ($0.934 \to 0.0033$, a $280\times$ reduction) while keeping the RMS within 0.038 px of the unregularised baseline. At α=10^{-3} the weak-mode drift is below 10^{-3} and the RMS penalty is only 0.038 px—the prior blocks the poorly observable directions while leaving the well-observed degrees of freedom free to improve the fit. At the strongest setting (α=10), the Schur prior still converges, while the isotropic prior produces a higher RMS (0.286 px vs 0.323 px) with residual weak drift (0.010).

3.9.3 The Double Role of the Rayfield

This result formalises a double role for the rayfield estimate θ_0 in direct bundle adjustment:

1. **Initialiser.** θ_0 places the direct BA in the correct convergence basin (Section 3.8), avoiding the local minima that trap a pinhole or perspective-CMO initialisation (RMS 29 px vs. 0.88 px).

2. **Observability prior.** The Schur eigenmodes of the Fisher matrix at θ_0 tell the optimiser *which directions it may trust*. The prior blocks compensation between poses and intrinsics without penalising the genuinely observable optical degrees of freedom.

The Schur prior is not a generic regulariser—it is *derived from the same data* that produced the initialisation. The rayfield's Fisher information matrix, evaluated at θ_0, encodes the local observability structure of the inverse problem. Using that structure to guide the subsequent BA is the methodological contribution: the rayfield is not merely a calibration model but an *observability oracle* for direct refinement.

4 Results

4.1 Data Sources

Table 3 clarifies which results come from real Pycaso calibration data, which from synthetic oracles, and which are methodological diagnostics.

Table 3. Data sources for results reported in this paper. Real data: 10 Pycaso stereo pairs. Synthetic oracles: StereoComplex `model_selection_oracles.py`. Method: analytical and numerical diagnostics independent of data source.

Result category	Source	Examples
Real data	Pycaso CMO images (10 stereo pairs)	Zernike fit (0.47 px), CMO+SE(3) (1.06 px), descriptors (b, WD, θ), pose sweep, cross-validation, bootstrap, f_x sensitivity
Synthetic oracles	StereoComplex simulator (noiseless)	Direct vs. rayfield inversion ($c = 0.81$), ray-space BIC, oracle parameter recovery
Methodological	Analytical / numerical diagnostics	BIC formula, Plücker residual decomposition, SE(3) ablation, Schur derivation

4.2 Dataset and Pipeline

Ten stereo pairs of 2048×2048 px ChArUco [9, 22] images (16×12 squares, 0.3 mm, DICT_6X6_250, `setLegacyPattern(True)`) from the Pycaso open dataset [8] are used. The Z-range is 2.65 mm–3.35 mm ($\Delta = 0.70$ mm). Corner detection yields 61–161 corners per frame; Hessian-based completion ($|\det H|$ + Otsu + barycentre) fills all 165. A double Ray2D TPS denoising pass [3] (ArUco markers at $\lambda = 10$, then completed 165 corners at $\lambda = 3$, Huber $c = 1.5$) produces gauge-stable 2D corners. The two-phase TPS is not cosmetic: without it, raw ChArUco detections carry structured biases from compression artifacts, blur, and defocus that cause the Zernike rayfield fit to become gauge-unstable—the direction piston Z_0 drifts by $8.5°$ between constrained and full-pose fits, making physical interpretation impossible. After double TPS, the gauge drift drops to $0.023°$, and the rayfield becomes a stable experimental oracle [33].

4.3 Pose Count Sensitivity

To assess robustness to the number of calibration frames, we sub-sample $N \in \{3, 5, 8, 10\}$ frames from the 10 available and re-run the full pipeline (Zernike fit + 26p model) for each. Table 4 reports pixel RMS and effective geometric descriptors.

Table 4. Pose sweep: pixel RMS and effective-descriptor stability vs. number of frames (all descriptors in the rayfield gauge with $f_x = 25600$ px). Each row is an independent refit on N frames; the N=10 value (1.07 px) reproduces the main-text calibration (1.06 px, full pipeline) to within refit noise.

N frames	RMS (px)	P50 (px)	b (mm)	WD (mm)	θ (°)
3	1.02	0.84	24.7	64.6	22.4
5	1.03	0.84	24.8	64.6	22.5
8	1.06	0.87	24.9	64.7	22.6
10	1.07	0.88	24.8	64.7	22.4

The pixel RMS is stable within 0.05 px across frame counts—even 3 frames suffice for 1.02 px calibration. Effective descriptors vary by less than $0.2\,\mathrm{mm}$ (baseline) and $0.1\,\mathrm{mm}$ (working distance) in the chosen gauge, confirming that the pipeline is robust to the number of calibration images. The effective convergence angle $\theta \approx 22.5°$ (standard deviation $0.1°$).

4.4 OpenCV Tuning Does Not Recover

We tested the full OpenCV stereo calibration flag space: `CALIB_RATIONAL_MODEL` (8 distortion coefficients), `CALIB_THIN_PRISM`, and `CALIB_TILTED_MODEL`. None of these configurations produced a usable calibration on the CMO data. The rational model, with the most parameters (14 intrinsics per camera), still fails because the distortion model remains radial and centred—it cannot represent a non-central projection where rays originate from two different sub-pupils. The best OpenCV configuration achieves $>300\,\mathrm{px}$ RMS, compared to $1.06\,\mathrm{px}$ for our 26-parameter CMO+SE(3) model. This confirms that the failure is structural (central projection assumption), not a matter of under-parameterised distortion.

4.5 Geometric Descriptor Consistency

The manufacturer documentation identifies the instrument as a Leica M205C common-main-objective (CMO) stereomicroscope, equipped with a Planapo 1.0× objective, an HD-V photo/video tube, a 1.0× video/photo objective and a 1.0× C-mount adapter. For the Planapo 1.0× configuration, Leica specifies a working distance of $61.5\,\mathrm{mm}$ [13]. The datasheet does *not*, however, disclose the effective CMO stereo baseline, the sub-pupil positions, or the convergence angle—the internal geometry of the shared-objective design. These quantities are recovered here from the Zernike rayfield rather than read from the documentation, which is precisely where the rayfield is informative.

A quantitative consistency check follows from the chief-ray geometry of the telecentric CMO model used in this work. The two stereo channels view the object through two off-axis sub-pupils separated by a baseline b in the exit-pupil plane of the shared objective. By the focal (Fourier) property of the objective, a pupil point at transverse height y maps to a chief ray that leaves object space at an angle θ_y with $\tan\theta_y = y/f_{\mathrm{obj}}$, where f_{obj} is the effective axial length (the sub-pupil-plane to working-plane distance in the CMO chief-ray model). The two sub-pupils at $\pm b/2$ therefore emit chief rays inclined at $\pm\arctan\big(b/(2f_{\mathrm{obj}})\big)$ to the optical axis, so the full convergence angle obeys

$$\Theta = 2\arctan\frac{b}{2f_{\mathrm{obj}}}, \qquad \text{equivalently} \qquad b = 2\,f_{\mathrm{obj}}\,\tan\frac{\Theta}{2}, \tag{15}$$

in which the relevant axial length is the sub-pupil-plane-to-working-plane distance $f_{\mathrm{obj}} = WD - z_p$ in the CMO chief-ray model; we do not require it to be interpreted as the manufacturer objective focal length.

The convergence angle $\Theta = \arccos(d_L \cdot d_R) = 22.6°$ is a *scale-invariant* rayfield observable—it depends only on the reconstructed ray directions, not on the chosen reference scale. Together with the reconstructed $b = 24.9\,\mathrm{mm}$, $WD = 64.7\,\mathrm{mm}$ and $z_p = 2.5\,\mathrm{mm}$ (hence $f_{\mathrm{obj}} = 62.2\,\mathrm{mm}$), the descriptors satisfy (15) internally: $2 \times 62.2\,\mathrm{mm} \times \tan(11.3°) = 24.9\,\mathrm{mm} = b$, confirming that the centre-pixel rayfield obeys the CMO chief-ray geometry. The *external* check uses the only absolute length the datasheet provides. Adopting the documented working distance as the axial scale, $f_{\mathrm{obj}} \approx WD_{\mathrm{doc}} = 61.5\,\mathrm{mm}$, and the measured Θ, relation (15) predicts

$$b_{\mathrm{pred}} = 2 \times 61.5\,\mathrm{mm} \times \tan(11.3°) = 24.6\,\mathrm{mm}, \tag{16}$$

within $0.3\,\mathrm{mm}$ ($\approx 1\,\%$) of the reconstructed $b = 24.9\,\mathrm{mm}$; equivalently, the reconstructed baseline and the documented working distance imply $\Theta_{\text{pred}} = 2\arctan\big(24.9/(2 \times 61.5)\big) = 22.9°$, against the measured $22.6°$. The blind, Zernike-informed reconstruction thus recovers a CMO geometry quantitatively compatible with the documented Planapo $1.0\times$ configuration.

This agreement is a strong physical *consistency* check, not an independent metrological validation of the Leica specifications. The convergence angle is scale-invariant, but the absolute lengths (b, WD, f_{obj}) inherit the chosen reference focal length $f_x = 25\,600\,\text{px}$, and WD scales linearly with it (§4.7); reproducing the documented WD to a few percent shows only that this scale choice is compatible with the documented optics, since it still relies on the datasheet WD as the scale anchor rather than on an independent 3D measurement. What the test does establish is that the rayfield recovers internal CMO parameters absent from the manufacturer documentation, and that they are quantitatively consistent with the documented Planapo $1.0\times$ configuration. Beyond this external check, the descriptors are also **internally consistent** by three independent tests: (1) cross-validation stability: 1.07 ± 0.02 px RMS across 5 folds, with effective-descriptor spread ± 0.4 mm (baseline), ± 0.15 mm (WD); (2) bootstrap 95 % confidence intervals: $b \in [23.48, 25.04]$ mm, $WD \in [64.64, 65.13]$ mm, $\theta \in [21.04°, 22.72°]$; (3) frame-count stability: effective descriptors vary by < 0.2 mm (baseline) and < 0.1 mm (WD) from 3 to 10 frames (Table 4).

An independent *absolute* validation—a certified 3D artefact at a known distance, or a direct measurement of the baseline or working distance—remains open and is listed among the limitations (Section 5.2). The present result nonetheless demonstrates the tool's value: from images alone, the rayfield recovers the undisclosed internal CMO geometry and places it in quantitative agreement with the manufacturer's documented Planapo $1.0\times$ configuration.

4.6 Comparison with the Soloff Polynomial Baseline

To calibrate the accuracy–interpretability trade-off, we benchmark against the Soloff polynomial model [28], a standard approach in stereo-microscope calibration [8]. A Soloff model maps 3D world coordinates (X, Y, Z) directly to pixel coordinates (u, v) via a polynomial expansion $u = \sum a_i \phi_i(X, Y, Z)$, $v = \sum b_i \phi_i(X, Y, Z)$ using monomials ϕ_i up to a chosen degree. Four independent polynomials are fitted (left-u, left-v, right-u, right-v) by ridge-regularised least squares on the same ChArUco corner observations used for our Zernike rayfield—identical input data, identical train/evaluation split. The model has no notion of 3D rays, no geometric consistency constraint between channels, and no optical interpretation of its coefficients.

Table 5 reports the pixel reprojection RMS for Soloff fits of degree 1 to 4 alongside the Zernike rayfield and the CMO 26p model.

Table 5. Pixel reprojection RMS on the Pycaso dataset: Soloff polynomial vs. Zernike rayfield vs. CMO 26p. All models are evaluated on the same 1,650 stereo corner correspondences (10 frames $\times$ 165 corners), i.e. 3,300 image observations over the two channels. Soloff fits are ridge-regularised ($\lambda = 10^{-3}$). The Zernike RMS floor (~ 0.41 px) is the best achievable under the 3D geometric consistency constraint; the Soloff floor (~ 0.18 px) is the best achievable by unconstrained 2D regression. The 0.23 px gap quantifies the cost of enforcing a physically meaningful 3D ray model.

Model	Params	RMS (px)	Interpretable
Soloff deg. 1	16	1.66 px	No
Soloff deg. 2	40	0.22 px	No
Soloff deg. 3	80	0.18 px	No
Soloff deg. 4	140	0.18 px	No
Zernike O(0)+d(2)	57	0.47 px	Rayfield
Zernike O(1)+d(3)	93	0.41 px	Rayfield
CMO telecentric 14p	14	14.6 px	Yes
CMO 26p (SE(3))	26	1.06 px	Yes

Three observations follow. First, the Soloff polynomial is a more accurate pixel regressor than any 3D-geometric model: at comparable parameter counts, Soloff deg. 2 (40 params, 0.22 px) outperforms Zernike O(0)+d(2) (57 params, 0.47 px) by a factor of $\sim 2.1\times$, and Soloff deg. 3 (80 params, 0.18 px) outperforms Zernike O(1)+d(3) (93 params, 0.41 px) by $\sim 2.3\times$. The polynomial reaches a noise floor of 0.18 px at degree 3, beyond which additional parameters (≥ 140) produce no further improvement.

Second, the Zernike rayfield reaches its own floor at 0.41 px, approximately 0.23 px above the Soloff floor. This gap quantifies, for this dataset and model family, the cost of enforcing 3D geometric consistency: the rayfield must produce a per-pixel 3D line whose intersections with rays from the other channel are geometrically meaningful, whereas Soloff merely interpolates 2D coordinates. The gap is not a failure of the Zernike basis—it is a physical consequence of solving a harder problem.

Third, the CMO 26p model loses an additional 0.65 px relative to the Zernike reference, but gains a physically interpretable parameterisation: its 26 parameters are the baseline, working distance, effective axial length, convergence angle, shear coefficients, and per-arm SE(3) poses. No Soloff coefficient can be read as a baseline or a working distance.

These figures are the model *floors at construction*, not the endpoint. The 1.06 px of the CMO 26p model is its rayfield-identified *initialisation*; once that initialisation seeds the free-pose Schur-stabilised direct bundle adjustment (§3.9), the same 26 parameters refine to 0.28 px, narrowing the gap to the Soloff degree-3 floor to under 0.1 px. After refinement, then, the compact physical model is competitive in pixel accuracy with the black-box regressor while retaining a physically interpretable parameterisation—and it does so without the optics–pose compensation that the unregularised free-pose optimum (0.24 px) exploits.

The comparison thus frames the paper's central trade-off explicitly. At construction: 0.18 px (black-box, no geometry), 0.41 px (geometric rayfield, no optical model), 1.06 px (compact physical model, physically interpretable parameterisation). After free-pose Schur-stabilised refinement the physical model closes most of that gap, reaching 0.28 px—so interpretability ultimately costs a fraction of a pixel rather than a category of accuracy.

Beyond the accuracy–interpretability trade-off, the field-resolved rayfield does what a diagnostic instrument should: it points to the next physical degree of freedom missing from the compact model.

The Zernike rayfield reveals that the real instrument is not perfectly telecentric: the effective sub-pupil positions shift laterally by approximately $1.0\,\mathrm{mm}$ across the field, producing a centre-to-edge baseline variation that the 26-parameter telecentric skeleton misses entirely (it holds the sub-pupils at fixed lateral positions). This residual non-telecentricity is a natural extension of the model; but the per-parameter identifiability analysis of §3.7 shows that any parameter controlling axial or lateral sub-pupil scale lies in the null space of the planar ChArUco observations and cannot be identified without an external reference. The specimen-level axial gain $\gamma_z \approx 0.70$ (§4.8) is consistent with this interpretation: it closes the CMO–Zernike relief gap in post-processing, but cannot be promoted to a calibrated model parameter from planar data alone. The rayfield thus fulfills its diagnostic role not by claiming completeness, but by identifying exactly which degree of freedom remains unresolved—and under what conditions it could be anchored.

4.6.1 Extrapolation to Held-Out Poses

The accuracy–interpretability trade-off above is measured on the full 10-frame dataset (train = test, since both models use all available data). A stricter question is: how well does each model *extrapolate* to poses not seen during fitting? This tests whether the 3D geometric constraint in the Zernike rayfield provides a regularisation benefit beyond what the Soloff polynomial's ridge penalty achieves.

We split the 10 frames into 7 training poses (the most central ones, closest to the centroid of the pose distribution) and 3 held-out test poses (the most extreme, at the edges of the sampled volume). Both models are fitted on the training set only and evaluated on the held-out frames. Critically, the two model families have different natural output spaces—Soloff maps 3D coordinates to pixels, Zernike maps pixels to 3D rays—so their extrapolation errors are measured in their respective native metrics and compared via the *degradation ratio* (test error / train error).

For Soloff, the native metric is the 2D pixel prediction error (the RMS distance between the polynomial-predicted pixel and the observed corner). For the Zernike rayfield, the native metric is the 3D triangulation accuracy: observed left and right pixels are converted to rays via the rayfield, the rays are intersected to obtain a 3D point, and this point is compared to the true corner position (known from the board model and the frame pose estimated by a Kabsch alignment of the triangulated points to the board). The Kabsch step removes the rigid gauge freedom inherent to uncalibrated stereo reconstruction (the absolute pose of the board cannot be recovered without an external reference), so the residual measures only the non-rigid rayfield quality.

Table 6 reports the results. Both models degrade by a factor of approximately $1.9\times$ on held-out extreme poses, confirming that the 7-frame training set is sufficient to build a usable model in both cases. The crucial difference appears at higher model complexity: Soloff degree 3 (80 parameters) degrades by a factor of $4.65\times$ (from $0.16\,\mathrm{px}$ to $0.75\,\mathrm{px}$): its extra high-order terms lower the in-sample residual but capture training-pose-specific structure that does not transfer to the held-out extreme poses—the operational signature of overfitting. We do not claim a single mechanism; the absorbed structure may be detection noise, pose-dependent systematic residuals, or high-order instability when extrapolating beyond the training pose distribution. The Zernike rayfield (180 parameters, O(4)+d(4)), by contrast, degrades by only $1.86\times$ despite having more than twice as many parameters. The 3D geometric consistency constraint—every pixel's ray must intersect consistently with rays from the other channel and with the board geometry—acts as a structural regulariser that prevents the rayfield from memorising individual training poses.

Table 6. Extrapolation to held-out extreme poses (7 train / 3 test). Each model is evaluated in its native metric: 2D pixel error for Soloff (3D→2D mapping), 3D triangulation accuracy for Zernike (2D→3D mapping). The degradation ratio (test / train) is the comparable quantity across model families.

Model	Params	Train	Test (extreme)	Degradation
Soloff deg. 1	16	1.21 px	2.48 px	2.05×
Soloff deg. 2	40	0.20 px	0.39 px	1.97×
Soloff deg. 3	80	0.16 px	0.75 px	**4.65**×
Zernike O(4)+d(4)	180	0.0006 mm	0.0012 mm	**1.86**×

Two conclusions follow. First, at comparable degradation ratios ($\sim 1.9\times$), Soloff remains more accurate in its native 2D pixel metric whereas the Zernike rayfield remains stable in its native 3D triangulation metric, consistent with the in-sample comparison of Table 5: the 2D polynomial is a more accurate regressor, but its accuracy collapses when the model order is pushed (degree 3). Second, the Zernike rayfield's degradation stays bounded ($1.86\times$) even at 180 parameters because the 3D geometric consistency constraint prevents it from overfitting to individual poses. The rayfield does not learn "pixel (500, 1300) corresponds to 3D point X in frame 3"; it learns "pixel (500, 1300) emits a ray that must intersect the corresponding ray from the other channel at a 3D point consistent with a rigid board". This constraint is absent from the per-coordinate polynomial regression, which treats each pixel coordinate as an independent scalar function of (X, Y, Z).

We deliberately do not convert the Zernike 3D error to a pixel-equivalent: that conversion factor depends on the local ray geometry (pixel pitch, working distance, ray obliquity) and the inverse map is, moreover, ill-conditioned at the microscope's magnification ($f_x = 25600$ px gives $\partial d/\partial u \approx 2.5\,\mu\text{m/px}$, so a sub-millimetre ray miss displaces the inverted pixel by tens of pixels). The degradation ratio of Table 6 sidesteps this by comparing each family in its native metric.

A complementary and more direct comparison expresses *both* models in a single physical metric—the 3D reconstruction error in micrometres. This requires a shared board-pose reference: the absolute board pose is a gauge freedom that neither uncalibrated model fixes on its own, and comparing the Zernike reconstruction against the CMO-model poses alone would conflate the rayfield error with the sub-millimetre disagreement between the two bundle adjustments' pose estimates (an unaligned comparison inflates the Zernike error to $\sim 500\,\mu\text{m}$, almost entirely pose mismatch). We therefore freeze the poses and evaluate two symmetric protocols. *(i) CMO-pose reference:* the Zernike rayfield is refitted with the board poses held fixed at the CMO-model values $\{R_i, t_i\}$—only the 180 rayfield coefficients are optimised—so both models share the exact pose set that the Soloff polynomial implicitly uses; a single global rigid (Kabsch) transform then removes the residual camera-to-world gauge of the triangulated points. *(ii) Zernike-pose reference:* conversely, the per-frame poses self-consistent with the free-fit Zernike field are recovered by pose-only least squares, and the four Soloff polynomials are refitted against that geometry. Reporting both protocols brackets the reference dependence and tests whether the conclusion survives the choice of which model defines the board geometry.

Table 7. 3D reconstruction RMS (micrometres) on held-out extreme poses, with both models expressed against a shared board-pose reference. Two symmetric protocols bracket the reference dependence: poses frozen to the CMO model, or to the free-fit Zernike field. A single global rigid (Kabsch) transform, estimated on the training frames, removes the camera-to-world gauge. The qualitative ranking is invariant to the reference choice.

		CMO-pose ref.		Zernike-pose ref.	
Model	Params	Train→Test (µm)	Degr.	Train→Test (µm)	Degr.
Soloff deg. 2	40	1.24 → 2.14	1.72×	2.58 → 4.78	1.86×
Soloff deg. 3	80	0.99 → 3.81	**3.86**×	1.99 → 3.87	1.95×
Zernike O(4)+d(4)	180	1.18 → 2.39	2.04×	2.63 → 5.53	2.10×

Table 7 shows the ranking is invariant to the reference choice. In both protocols the 180-parameter Zernike rayfield *matches the accuracy of the 40-parameter Soloff degree-2 fit* ($1.2\,\mu\mathrm{m}/2.4\,\mu\mathrm{m}$ train/test under the CMO reference, $2.6\,\mu\mathrm{m}/5.5\,\mu\mathrm{m}$ under the Zernike reference) and degrades by only $\sim 2\times$, never collapsing. Soloff degree 3 attains the lowest training error in both references, but its degradation is reference-dependent: against the cleaner CMO geometry (the CMO model fits all ten frames jointly) it overfits sharply ($3.86\times$), consistent with its native pixel-metric collapse ($4.65\times$, Table 6); against the noisier Zernike-recovered geometry the same overfit is partly masked ($1.95\times$) because the $\sim 2\,\mu\mathrm{m}$ pose-recovery floor inflates the training residual. The absolute micrometre values and the degree-3 overfit magnitude therefore depend on which model's poses define the reference—neither constitutes an external metrological truth, consistent with the absolute-scale caveats above—but the qualitative conclusion is robust across both: the Zernike rayfield buys 3D geometric consistency and overfit resistance at the cost of $\sim 0.2\,\mathrm{px}$, whereas a high-order Soloff polynomial maximises raw pixel accuracy at the risk of overfitting.

4.7 Internal Validation

Three internal validation experiments assess the robustness of the 26p pipeline on the Pycaso dataset.

Cross-validation. 5-fold leave-2-out cross-validation (fit on 8 frames, evaluate on 2 held-out frames) yields a mean pixel RMS of 1.07 ± 0.02 px. The low standard deviation ($\sim 0.02\,\mathrm{px}$) confirms that the model is not overfitting to specific frames; the calibration is stable across the 10-frame dataset. Fold 1 shows slightly higher RMS ($1.09\,\mathrm{px}$) and a lower effective baseline estimate ($b = 23.1\,\mathrm{mm}$) because the held-out frames (0 and 1) cover the most extreme Z values.

Bootstrap descriptor stability. 100 bootstrap iterations (resampling 10 frames with replacement, refitting the full pipeline each time) give 95 % confidence intervals:

- Effective baseline b: [23.48, 25.04] mm (mean 24.56 mm, $\sigma = 0.43$ mm)
- Effective working distance WD: [64.64, 65.13] mm (mean 64.86 mm, $\sigma = 0.15$ mm)
- Effective convergence angle θ: [21.04°, 22.72°] (mean 22.17°, $\sigma = 0.46°$)

The effective working distance is the most stable descriptor ($\sigma < 0.2\,\mathrm{mm}$), while the effective baseline and convergence angle show $0.4\,\mathrm{mm}$ and $0.4°$ variation respectively—consistent with the gauge sensitivity of the Zernike origin field noted in Section 5.2.

Sensitivity to fixed focal length f_x. Varying the fixed pinhole focal length f_x by $\pm 10\%$ (5 values: 23040, 24320, 25600, 26880, 28160 px) produces:

- Pixel RMS: $1.21 \rightarrow 0.95$ px—remains near the one-pixel range, confirming that the rayfield is stable as a calibration object.
- Working distance: $58.3 \rightarrow 71.3$ mm—linearly proportional to the initial fx ($\pm 10\%$ fx $\rightarrow \pm 10\%$ WD). WD is directly scaled by the pinhole reference and is *not* independently identifiable.
- Convergence angle: $24.1° \rightarrow 20.5°$—varies inversely with WD.
- Baseline: $23.9 \rightarrow 25.1$ mm ($\pm 2.5\%$)—moderately stable.

The absolute physical descriptors, especially WD and θ, remain sensitive to the chosen pinhole reference scale. Descriptor readout therefore requires either an independently known optical scale (pixel pitch, magnification) or external metrological validation—a known limitation of all single-view calibration methods without a known reference object.

4.8 Application Sanity Check: Specimen Reconstruction

A non-metrological sanity check validates the calibrated rayfields on an independent speckled coin specimen imaged with the same Pycaso microscope (single stereo pair, no ChArUco pattern). This is **not** absolute metrological validation—no ground-truth 3D data exists for the coin—but it confirms that the rayfields produce coherent dense reconstruction on data unseen during calibration. Dense left-to-right pixel correspondences are computed with DIS optical flow over a 1448×1448 px ROI; within the 1400×1400 px analysis window shown in Figure 9, $95.4\,\%$ of pixels carry a valid correspondence. For each correspondence, the two rays from the calibrated model are intersected at their closest point (ray-gap triangulation).

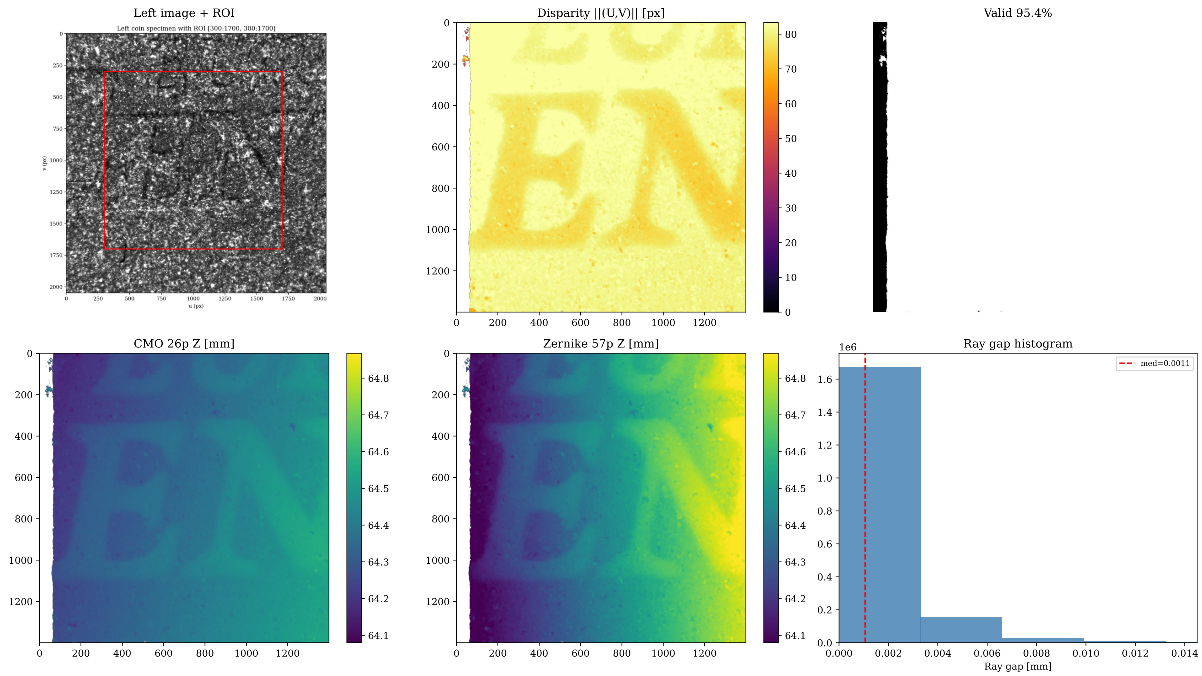

Figure 9. Dense specimen reconstruction. Top row: left image, DIS optical-flow magnitude $\|(U, V)\|$, valid mask ($95.4\,\%$ coverage over the displayed window). Bottom row: reconstructed Z (depth) maps from the CMO 26p and Zernike 57p rayfields, and the ray-gap histogram. Both BA-refined models produce nearly identical ray gaps (median 0.001 mm), confirming rayfield coherence. Quantitative surface roughness and ray-gap values for all five model variants (rayfield-initialised through Schur-regularised) are reported in Table 8.

The median ray gap is $0.001\,\mathrm{mm}$ for both BA-refined models in Figure 9—excellent coherence, confirming that the two channels' rays intersect each other at the sub-millimetre level, despite never having been jointly optimised on this specimen. The CMO 26p reconstructions are smoother than the Zernike rayfield reconstruction because the compact physical model imposes implicit spatial smoothness through its rigid-sub-pupil and affine-direction parameterisation, whereas the 57-parameter Zernike model has the flexibility to absorb detection noise into high-frequency surface variations. Quantitative Z MAD and ray-gap values for the rayfield-initialised, unregularised BA, isotropic-prior BA, and Schur-prior BA variants are reported in Table 8. OpenCV stereo reconstruction was not available because the standard OpenCV calibration failed to converge on this microscope's calibration data.

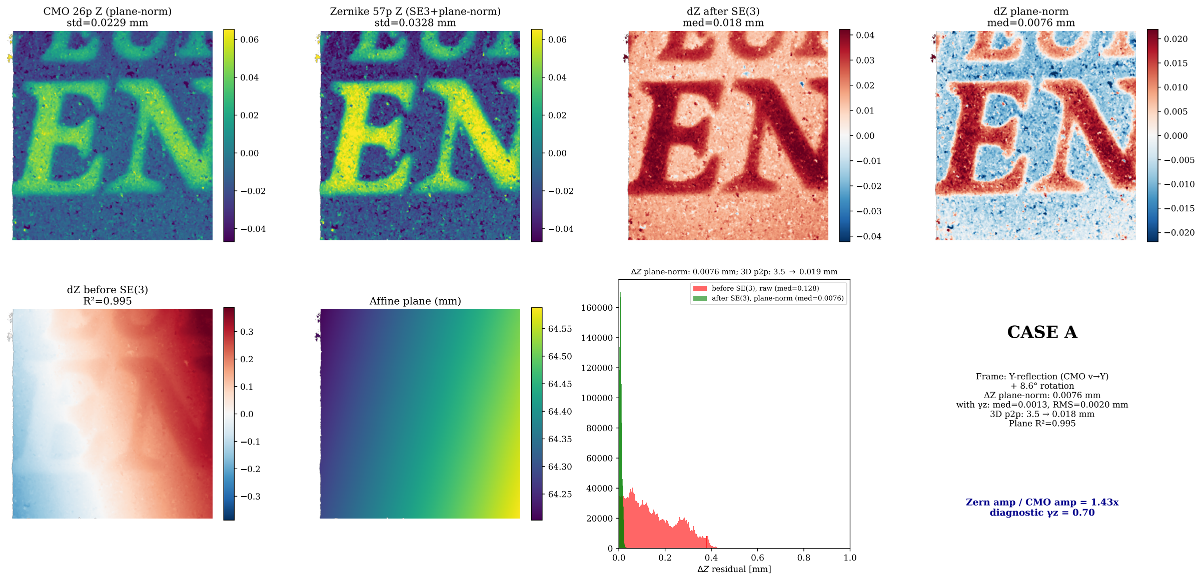

Figure 10. Rigid-gauge removal and axial-gain diagnostic: Zernike vs CMO surface comparison. Top row: CMO and Zernike surfaces after plane-normalisation (relief only), followed by ΔZ before and after plane normalisation. Bottom row: ΔZ before SE(3) alignment (dominated by an affine ramp, $R^2 = 0.995$), the fitted affine plane, and the residual histogram. After SE(3) alignment the median plane-normalised ΔZ residual is $0.0076\,\mathrm{mm}$; the corresponding median 3D point-to-point residual drops from $3.5\,\mathrm{mm}$ to $0.018\,\mathrm{mm}$ (a $99.5\,\%$ reduction). The remaining relief-amplitude ratio is scalar: $Z_{\mathrm{Zernike}}/Z_{\mathrm{CMO}} = 1.43\times$, i.e. $\gamma_z \approx 0.70$. Rescaling the CMO relief by $1/\gamma_z$ reduces the CMO–Zernike residual to $0.0013\,\mathrm{mm}$ median absolute ΔZ ($0.0020\,\mathrm{mm}$ RMS). Thus the residual is not a spatially arbitrary non-rigid discrepancy; it is mainly an axial stereo-depth gain.

The specimen comparison is intentionally presented in two steps, because the apparent disagreement is scientifically useful. First, the SE(3) alignment shows that both models reconstruct the same surface up to the CMO model's $v \to Y$ sign convention (a Y-axis reflection) and a small ($\sim 8.6°$) rigid rotation. Second, after these gauge effects are removed, a residual depth-relief discrepancy remains: the Zernike reconstruction exhibits approximately $43\,\%$ larger relief amplitude than the CMO 26p model (standard deviation ratio $1.43\times$, Figure 10). This is the diagnostic observation that motivates the axial-gain analysis below.

To diagnose whether the discrepancy is a uniform scale error, a non-rigid shape error, or a depth-only effect, we fit nested transformations from Zernike to CMO points on matched specimen correspondences. An SE(3)-only fit (no scale) yields a median residual of $0.009\,\mathrm{mm}$, already confirming excellent geometric agreement. This $0.009\,\mathrm{mm}$ is the SE(3)-only fit residual on the $50\,000$ subsampled correspondences, distinct from the $0.018\,\mathrm{mm}$ dense-map median 3D point-to-point residual reported in Figure 10. Adding a global isotropic scale (Sim(3)) produces an optimal scale factor $s = 1.00$, ruling out a uniform 3D magnification—the difference is not a simple focal-length or pixel-scale error. However, an anisotropic scale model with independent (s_x, s_y, s_z) recovers $s_x \approx s_y \approx 1.00$ but $s_z \approx 0.70$, indicating a **depth-only scale difference** of approximately $30\,\%$ (consistent with the $1.43\times$ relief-amplitude ratio above). Equivalently, if the CMO 26p model is interpreted as predicting an ideal local depth–disparity sensitivity G_z^{CMO}, the specimen comparison suggests a diagnostic axial stereo gain

$$G_z = \gamma_z G_z^{\mathrm{CMO}}, \qquad \gamma_z \simeq \frac{\mathrm{std}(Z^{\mathrm{relief}}_{\mathrm{CMO}})}{\mathrm{std}(Z^{\mathrm{relief}}_{\mathrm{Zernike}})} \approx 0.70.$$

With this convention, $\gamma_z < 1$ means that the CMO 26p skeleton overestimates the depth–disparity sensitivity, so its reconstructed relief is too small by the reciprocal factor $\sim 1/\gamma_z$. The lateral dimensions (x, y) are identical between the two reconstructions; only the reconstructed depth range differs. Applying this single axial gain closes the internal CMO–Zernike discrepancy: the plane-normalised residual decreases from median $0.0076\,\mathrm{mm}$ / RMS $0.0100\,\mathrm{mm}$ / P95 $0.0182\,\mathrm{mm}$ to median $0.0013\,\mathrm{mm}$ / RMS $0.0020\,\mathrm{mm}$ / P95 $0.0037\,\mathrm{mm}$. In other words, after rigid-gauge removal and axial-gain correction, the compact CMO reconstruction falls back onto the Zernike relief to within about $2\,\mathrm{\mu m}$ RMS on this specimen.

This depth-only scale discrepancy is therefore not interpreted as a contradiction of the CMO family identification. It suggests that the compact 26-parameter skeleton captures the central chief-ray geometry, lateral magnification, and dominant CMO topology, but still ties the lateral magnification and the axial stereo-depth gain through a single idealised CMO relation. A real CMO microscope, with a common objective, zoom optics, tube optics, prisms, adapters, and finite pupil magnification, may have an effective axial stereo magnification that is not fully determined by the central baseline and convergence readout. This motivates a future augmented *CMO27* model with one explicit axial stereo gain parameter, implemented in ray space by the affine object-space gauge

$$T_{\gamma_z}(X, Y, Z) = (X, Y, \gamma_z Z), \qquad O' = T_{\gamma_z} O, \quad d' = \frac{T_{\gamma_z} d}{\|T_{\gamma_z} d\|}.$$

The CMO model's rigid sub-pupil and telecentric direction parameterisation impose a specific stereo geometry that the Zernike model, being a free rayfield, is not obligated to reproduce exactly. Since no independent 3D ground truth is available for the coin specimen, we interpret the depth-scale difference as a convergence-depth ambiguity inherent to uncalibrated stereo reconstruction, not as a failure of either model. The axial correction establishes *internal* CMO–Zernike consistency, not external metrological truth: it does not by itself establish which reconstruction is closer to the real coin surface, and γ_z should not be fitted freely without an axial reference because it would otherwise reopen the depth gauge. Resolving this point would require an independent 3D reference, for example known stage-Z steps or profilometry of the same specimen.

Table 8 quantifies the five reconstructions. The rayfield-initialised CMO model (before BA) has a median ray gap of $0.022\,\mathrm{mm}$—an order of magnitude worse than all BA-refined variants ($\sim 0.001\,\mathrm{mm}$). After enforcing a consistent world-frame convention, the initial model's larger triangulation gap becomes apparent: its triangulation quality is substantially worse than the BA-refined models. All BA variants recover tight ray intersections (median gap $\sim 1\,\mathrm{\mu m}$), confirming that the ray geometry is correct regardless

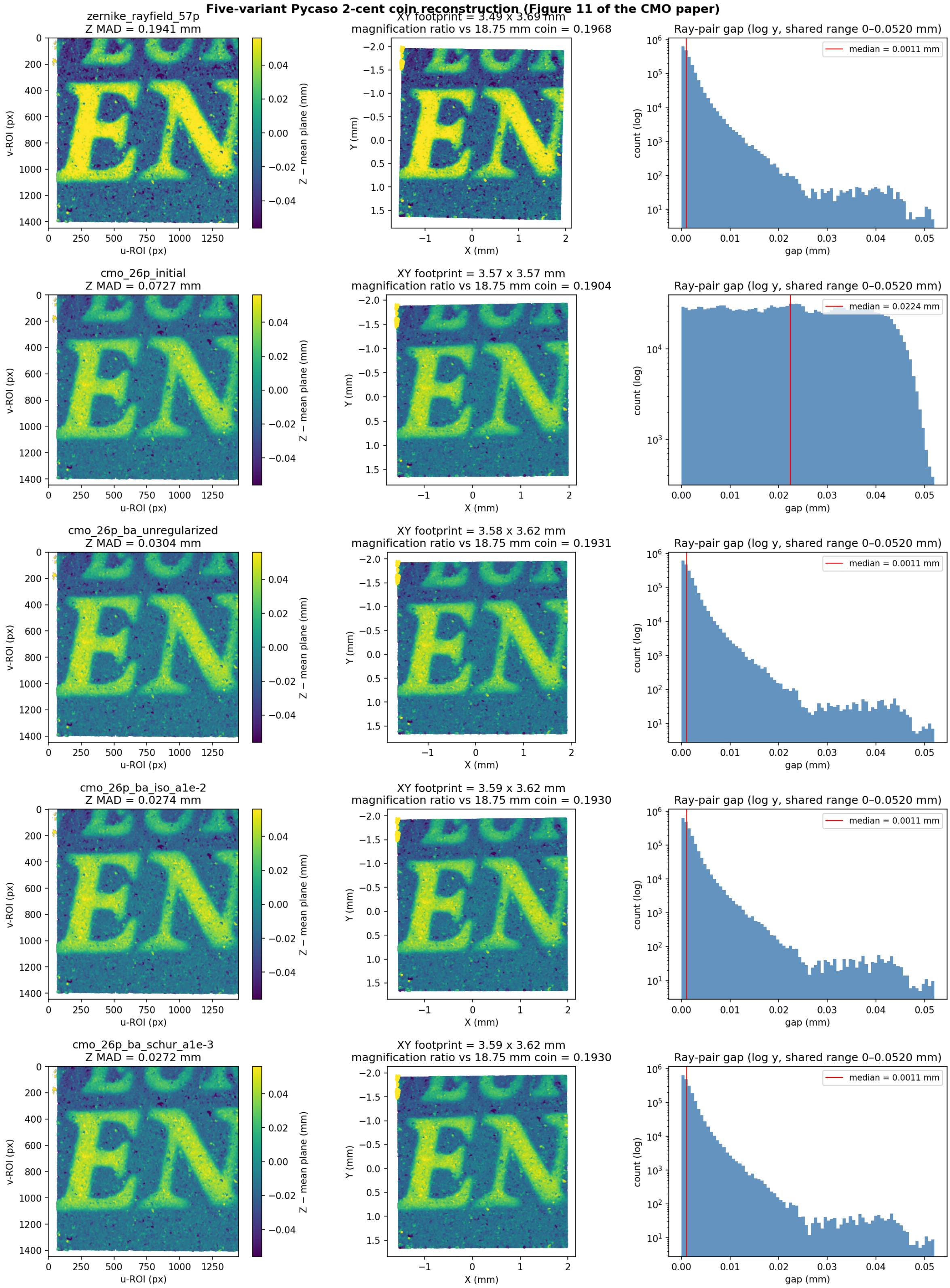

Figure 11. Five-variant specimen reconstruction with Schur-regularised variants. Each row shows the Z-map (surface relief after mean-plane subtraction), the XY footprint (coloured by relief), and the ray-gap histogram (log scale, shared range). All CMO variants are aligned to the Zernike reference frame. The rayfield initialisation (row 2) has a median gap $0.022\,\mathrm{mm}$—$20\times$ larger than the BA-refined rows—apparent once the world frame is made consistent. The regularised variants (rows 4–5) achieve Z MAD $0.027\,\mathrm{mm}$ (computed on the plane-normalised relief over the common valid ROI), marginally better than the unregularised BA ($0.030\,\mathrm{mm}$), $2.7\times$ smoother than the rayfield-initialised CMO variant ($0.073\,\mathrm{mm}$) and $7.2\times$ smoother than the Zernike map ($0.194\,\mathrm{mm}$); these MAD values are smoothness indicators, not metrological ground truth. All BA variants converge to magnification $\sim$0.193 vs. Zernike 0.197 (3.5 % systematic model bias)—a *lateral*-magnification difference, distinct from the depth-only axial gain $\gamma_z \approx 0.70$ diagnosed in Figure 10.

of the regularisation strategy. The regularised variants achieve Z MAD $0.027\,\mathrm{mm}$, marginally better than the unregularised BA ($0.030\,\mathrm{mm}$)—the prior does not degrade the geometric reconstruction. All CMO BA variants converge to a magnification ratio of $\sim$0.193 vs. the Zernike reference at 0.197, reflecting a 3.5 % lateral model bias between the compact 26-parameter physical model and the flexible 57-parameter rayfield. This lateral bias is small compared with the axial relief gain $\gamma_z \approx 0.70$ and, importantly, cannot explain it: the specimen diagnostic separates lateral magnification from depth/disparity sensitivity.

Table 8. Five-variant specimen reconstruction: surface roughness (Z MAD), median ray gap, and magnification ratio vs. the nominal 18.75 mm 2-cent euro coin.

Model	Z MAD (mm)	Median gap (mm)	Magnification
Zernike rayfield (57 p)	0.194	0.0011	0.1968
CMO 26 p (rayfield init)	0.073	**0.0224**	0.1904
CMO 26 p — BA unregularised	0.030	0.0011	0.1931
CMO 26 p — BA + isotropic prior (α=10^{-2})	0.027	0.0011	0.1930
CMO 26 p — BA + **Schur prior** (α=10^{-3})	**0.027**	0.0011	0.1930

4.9 Model Performance

Table 9. Model comparison on Pycaso CMO data. The 26-parameter CMO+SE(3) model is the compact physical reference. All rows use the operational constrained-pose protocol; the free-pose Schur-stabilised BA refines the same 26-parameter model to $0.28\,\mathrm{px}$ (Table 2), a diagnostic setting not listed here.

Model	p	Ray RMS (mm)	Px RMS (px)	P50 (px)	P95 (px)
OpenCV stereo	–	–	> 300	–	–
Perspective CMO	19	3.48	$\sim$86	–	–
Telecentric CMO + shear	14	0.12	14.6	13.2	22.4
CMO + SE(3)	**26**	**0.0021**	**1.06**	**0.87**	**1.84**
CMO + SE(3) + corner BA	26	$\sim$0.0019	0.88	0.64	1.70
Zernike O(0)+d(2)	57	0.0007	0.47	0.34	0.86

The 26-parameter CMO+SE(3) model achieves $1.06\,\mathrm{px}$ reprojection with less than half the parameters of the Zernike reference. It has a physically interpretable parameterisation: sub-pupils, effective axial length, telecentric slopes, and SE(3) arm corrections.

4.10 BIC Model Selection

Table 10. BIC model selection on the Zernike rayfield. Models above 1.5 px reprojection are REJECTED by the operational guard. The central and parallel-plate models have multi-millimetre ray RMS and are rejected on reprojection alone; their observability-penalised score S_{usable} then diverges ($\gg 0$) and is not tabulated precisely.

Model	p	RMS (mm)	$\mathrm{BIC}_{\mathrm{ray}}$	S_{usable}	Status
cmo telecentric shear	14	0.111	-36 128	+978 890	REJECTED
cmo telecentric	12	0.146	-33 201	+986 044	REJECTED
pinhole parallel plate	6	6.337	5 079	$\gg 0$	REJECTED
central pinhole	0	8.343	6 500	$\gg 0$	REJECTED
central brown conrady	10	8.324	6 549	$\gg 0$	REJECTED
CMO + SE(3) 26p	26	0.002	$-32\,433$	$-32\,433$	**BEST USABLE**

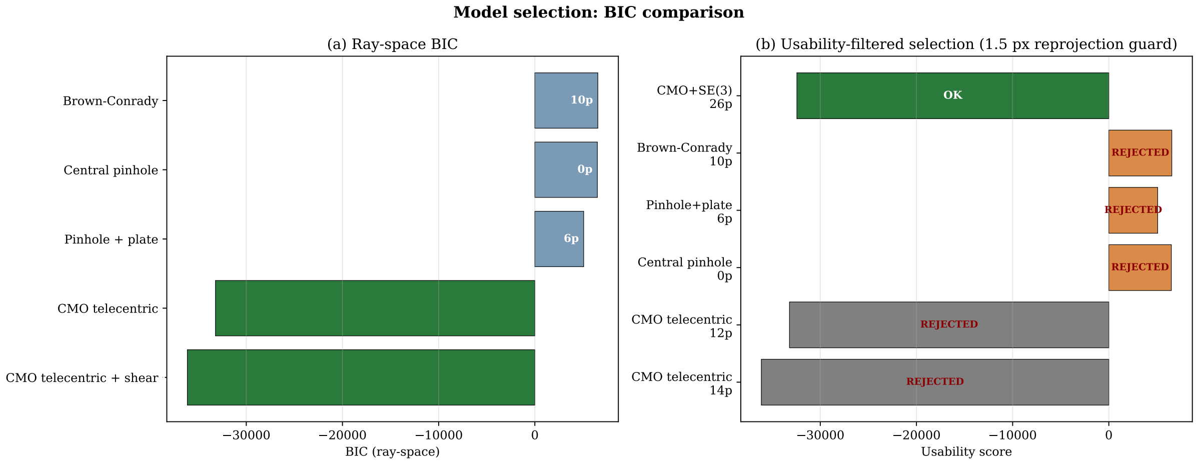

Figure 12. Two-panel BIC bar chart. (a) Ray-space BIC: the CMO telecentric family achieves the lowest BIC by a decisive margin ($> 40\,000$ points). (b) Operational usability score with a $1.5\,\text{px}$ reprojection guard: models exceeding the threshold are marked as REJECTED (shown in grey/orange); only the 26p SE(3)-aligned model passes the guard (shown in green).

The ray-space BIC provides decisive rejection of the non-CMO families ($\Delta\text{BIC} > 40\,000$), confirming that pinhole, Brown-Conrady, and parallel-plate models are grossly misspecified for this non-central system. The BIC magnitudes should be interpreted as rejection of families, not as certification of the retained model: the rayfield residuals are spatially correlated (the Zernike field is smooth), so the effective number of independent observations is lower than $N_{\text{obs}} = 3300$, inflating the absolute BIC scale. The ranking across families is robust to this effect. The operational usability score correctly rejects the 14p base model as unusable and selects the 26p aligned model.

4.11 Geometric Descriptors

Table 11. CMO-consistent effective geometric descriptors read directly from the Zernike rayfield at the centre pixel. All values are reported in the chosen rayfield gauge ($O \cdot d = 0$) and fixed reference focal scale ($f_x = 25600$ px); their absolute metrological interpretation requires an independent external reference (see Section 5.2).

Descriptor	Symbol	Value
Stereo baseline	b	$24.9\,\text{mm}$
Sub-pupil depth	z_p	$2.5\,\text{mm}$
Working distance	WD	$64.7\,\text{mm}$
Effective axial length	f_{obj}	$62.2\,\text{mm}$
Convergence angle	θ	$22.6°$

4.12 SE(3) Ablation

Table 12. SE(3) arm alignment ablation. Per-arm DOFs are individually necessary.

Variant	Params	Ray RMS (mm)	Px RMS (px)	P50 (px)	P95 (px)
Telecentric baseline	14	0.12	14.6	13.2	22.4
+ Rotation only	20	0.014	3.74	2.68	7.13
+ Translation only	20	0.010	2.44	2.06	4.15
+ Full SE(3)	**26**	**0.0021**	**1.06**	**0.87**	**1.84**
+ Shared rotation	23	0.0083	–	–	–
+ Shared translation	23	0.0041	–	–	–
+ Differential only	20	0.0070	4.04	2.85	7.74

Both rotation and translation are essential; neither alone suffices. Per-arm DOFs are individually necessary—any compression degrades the fit ($+92\,\%$ to $+289\,\%$ ray RMS increase).

5 Discussion

5.1 The Rayfield as Diagnostic Instrument

The key methodological contribution is the feedback loop: Ray2D preprocessing $\rightarrow$ Ray3D measurement $\rightarrow$ residual modal analysis $\rightarrow$ hypothesis $\rightarrow$ model refinement $\rightarrow$ verification. Without the rayfield, the 2D reprojection error tells you *that* the model is wrong but not *how*. The Zernike projection of Δd and Δm directly identifies whether the missing DOF is global (low-order Zernike modes) or spatial (high-order modes).

Figure 6 illustrates this diagnostic value concretely. The residual heatmap at each model stage is a **visual hypothesis generator**: the perspective CMO's structured d_y error (23.6° RMS) sends the user to telecentric models; the telecentric model's uniform Z_0^0 piston (0.27° RMS) sends the user to global arm alignment; and the SE(3)-refined model's near-zero residual (0.003° RMS) confirms convergence. At no point does the user need to guess—the residual pattern at each stage dictates the next hypothesis.

In this case study, the Z_0^0-dominated residual correctly guided us to the SE(3) arm alignment, while alternative hypotheses (pre-warp, variable origin) were correctly rejected by the same diagnostic.

5.2 Limitations

1. **Single dataset.** Results are demonstrated on 10 stereo pairs from one Pycaso CMO microscope. Generalisation to other CMO designs, manufacturers, or magnification settings has not been demonstrated and requires additional validation.

2. **No independent 3D ground truth.** All residuals (ray-space and pixel-equivalent) are computed on the same ChArUco board points used for calibration. There is no validation against a separate certified 3D object or an independently measured reference.

3. **Depth-scale ambiguity.** The specimen sanity check (§4.8) reveals a $\sim 30\,\%$ depth-scale difference between the Zernike and CMO 26p reconstructions ($\gamma_z \approx 0.70$). Applying the reciprocal axial gain makes the CMO relief agree with the Zernike relief to $0.0020\,\mathrm{mm}$ RMS, so the residual is internally explained by a single missing axial stereo-depth gain rather than by an arbitrary non-rigid error.

This still does not validate absolute depth: identifying γ_z metrologically requires an independent axial reference such as known stage-Z steps or profilometry of the same specimen.

4. **Effective rayfield descriptors are gauge-sensitive.** The geometric descriptors (b, WD, f_{obj}, θ) are read from the rayfield under the transverse gauge $O \cdot d = 0$ and a fixed pinhole reference $f_x = 25600$. The working distance is linearly proportional to the initial f_x ($\pm 10\%$ fx $\rightarrow \pm 10\%$ WD, Section 4.7), and is *not* independently identifiable without an external scale reference.

5. **SE(3) translation parameters are not uniquely identifiable.** The rotation angles ($\sim 2.5°$, $\sim 3.7°$) are stable across optimisation runs, but the translation components trade off with the telecentric base parameters (WD, f_{obj}, b, principal point). Only the SE(3) rotation is a robust diagnostic; the translation is effective rather than absolute.

6. **The Zernike rayfield is a training residual.** Its $0.0007\,\text{mm}$ ray-space RMS is a self-evaluation on the support points it was fitted to; the $0.47\,\text{px}$ pixel RMS is a noise-floor estimate, not an independent measurement. The pixel RMS column in all tables provides the apples-to-apples comparison.

7. **Fixed pinhole reference.** The Zernike BA uses a fixed $f_x = 25600$ and principal point at image centre. Errors in this reference propagate to the rayfield estimates and the derived physical descriptors.

8. **Absolute metrological validation would require:** (a) manufacturer specifications for the actual microscope used, (b) an independent measurement of WD or baseline, or (c) a certified 3D calibration object at a known distance. Item (a) is now partly available—the instrument is identified as a Leica M205C with a Planapo 1.0× objective (documented $WD = 61.5\,\text{mm}$ [13]), and the reconstructed baseline agrees with it to $\approx 1\,\%$ via the chief-ray relation (15) (§4.5)—but this is a physical *consistency* check that still uses the datasheet WD as the scale anchor, not an independent measurement; items (b) and (c) remain unavailable for the open Pycaso dataset. The present validation thus establishes internal geometric consistency, agreement with the documented optical configuration, model interpretability, and descriptor stability on a real CMO dataset, but not absolute 3D metrological accuracy. A future validation on a calibrated 3D artefact or stepped depth standard will be required to quantify absolute depth accuracy and uncertainty propagation.

5.3 Generalisability

The methodology—measure with a flexible non-parametric basis, read physical descriptors, diagnose residual structure, build compact physical model, validate by BIC—is transferable to any non-standard optical instrument (endoscopes, Scheimpflug cameras, multi-camera arrays). The SE(3) arm alignment is specific to CMO microscopes, but the residual analysis that identified it is general.

5.4 Why the Two-Stage Decomposition Works

The central insight of this work is not a specific model architecture but a problem decomposition strategy. Direct bundle adjustment jointly estimates optical parameters θ and board poses η from 2D corners; for non-central optics, the Schur complement of the reduced camera system [33] reveals coupling norms of $c = 0.81$–0.98 depending on the diagnostic configuration, meaning optics and poses are nearly inseparable—the problem is severely ill-conditioned. By inserting the Zernike rayfield as an intermediate variable, the estimation splits into two sub-problems: (1) fit a flexible rayfield to corners with a constrained pose model, and (2) identify physical optics in ray space, where the coupling norm is 0.0 by construction because board poses are no longer estimated (pose-free model selection with BIC).

This decomposition is the mechanism behind all three empirical findings: the TPS pre-processing stabilises step (1) by cleaning the 2D input; the rayfield decouples step (2) from pose estimation; and the rayfield-initialised parameters lie close to the constrained-pose corner optimum, while also providing the correct basin and observability prior for the free-pose Schur-stabilised refinement.

6 Conclusion

We have demonstrated the first real-data validation of the StereoComplex non-central calibration pipeline on a CMO stereo microscope. The key results are:

- A compact 26-parameter physical model (quasi-telecentric CMO skeleton + per-channel SE(3) arm alignment) achieves $1.06\,\mathrm{px}$ reprojection (P50=$0.87\,\mathrm{px}$, P95=$1.84\,\mathrm{px}$) where standard OpenCV stereo calibration fails (> 300 px).
- The Zernike rayfield serves as a diagnostic instrument: residual modal analysis identifies missing degrees of freedom, and the operational usability score with reprojection guard correctly selects the usable model.
- Effective rayfield descriptors—baseline, working distance, effective axial length, convergence angle—are read directly from the measured rays (within the chosen rayfield gauge and reference scale), without fitting a physical model.
- On the independent coin specimen, the remaining CMO–Zernike relief discrepancy is explained by a single axial stereo-depth gain $\gamma_z \approx 0.70$: applying this gain reduces the CMO–Zernike plane-normalised relief residual to $0.0020\,\mathrm{mm}$ RMS. This motivates a CMO27 extension while preserving the need for external axial ground truth.
- The rayfield-initialised parameters are near-optimal for direct corner bundle adjustment ($17\,\%$ improvement), and the same rayfield provides a Schur-complement observability prior that stabilises the BA against optics–pose compensation.
- The accuracy–interpretability trade-off is quantified by comparison with the Soloff polynomial baseline (§4.6): a black-box polynomial achieves $0.18\,\mathrm{px}$ (80 params) but yields no physical descriptors; the Zernike rayfield reaches $0.41\,\mathrm{px}$ (93 params) under the 3D geometric consistency constraint ($0.23\,\mathrm{px}$ gap = cost of geometry); the compact CMO 26p model is identified at $1.06\,\mathrm{px}$ and, used as the initialisation for the free-pose Schur-stabilised bundle adjustment, refines to $0.28\,\mathrm{px}$ (within $0.1\,\mathrm{px}$ of the Soloff floor) with a physically interpretable parameterisation. We therefore do not merely trade accuracy for interpretability: the refined physical model is competitive in pixel accuracy with black-box regression while, unlike it, producing a physically meaningful calibration.

The CMO Pycaso case study validates a generalisable methodology—measure first, model second, diagnose by residual, refine by hypothesis testing—and establishes the Zernike rayfield as a multi-purpose instrument: it measures the rays, diagnoses the missing physical degree of freedom, initialises the bundle adjustment, and regularises it in two distinct senses—*structurally*, its 3D ray constraint resists overfitting and extrapolates better than an unconstrained polynomial (§4.6), and *statistically*, its Schur-complement spectrum supplies an observability prior that suppresses optics–pose compensation (§3.9). Code, data, and documentation are publicly available at `https://github.com/jeffwitz/StereoComplex`. A self-contained Zenodo archive (concept DOI 10.5281/zenodo.20444215) bundles the manuscript, figures, tables, numerical audit, reproducibility scripts, and all data files needed to rebuild the PDF and regenerate every figure. For exact reproducibility, the version submitted with this manuscript is archived as v4 (10.5281/zenodo.20533009); the concept DOI above is a stable landing page that always resolves to the latest version.

Prospective

Several directions extend this work. First, the methodology should be validated on additional CMO microscope designs and manufacturers to confirm generalisability beyond the single Pycaso instrument. Second, the Greenough stereo microscope family—with tilted optical axes rather than parallel sub-pupils—presents a harder identification problem where the telecentric model does not apply; the residual-guided construction method would need to discover a different geometric skeleton. Third, quantitative uncertainty propagation from ChArUco corner noise to effective-descriptor estimates (baseline, working distance) would strengthen the metrological claim. Fourth, deep learning-based corner detection could replace the hand-tuned Hessian+TPS pipeline, potentially improving the raw 2D input quality and further reducing the gauge instability. Finally, extending the Zernike rayfield to time-varying deformations (e.g., vibration, thermal drift) would enable *in-situ* calibration monitoring without dedicated targets.

Reproducibility Statement

The StereoComplex codebase (commit `08d1b25`, branch `develop`) and all artefacts required to reproduce these results are available at `https://github.com/jeffwitz/StereoComplex`. The raw Pycaso calibration images are available from `https://github.com/LaboratoireMecaniqueLille/Pycaso`. To reproduce the numerical results without access to the raw images, restart from the pre-computed intermediate state (`intermediate_state.npz`) which contains the already-detected, Hessian-completed, and TPS-denoised corner positions, the fitted Zernike rayfield, and the initial 26p model parameters. A self-contained Zenodo archive provides the manuscript, all figures, tables, numerical audit, reproducibility scripts, figure-generation manifests, and key data files (`intermediate_state.npz`, specimen correspondences, Schur diagnostic). For exact reproducibility, the version submitted with this manuscript is archived as v4 (10.5281/zenodo.20533009, 100 files); the concept DOI (10.5281/zenodo.20444215) is a stable landing page under which future versions are deposited.

A Numerical Reprojection for the CMO Model

The CMO telecentric model defines a forward mapping $\mathtt{ray}(u, v) \to (O, d)$ from pixel coordinates to a 3D ray. Computing the 2D reprojection error requires the inverse: given a 3D point X^{cam} in the camera frame, find the pixel (u, v) whose ray passes through (or closest to) X^{cam}. For a non-central model, no closed-form inverse exists. We solve it numerically: for each observation, the predicted pixel is

$$(u_{\text{pred}}, v_{\text{pred}}) = \operatorname{argmin}_{(u,v)\,\in\,[0,W]\times[0,H]} \|X^{\text{cam}} - P(u,v;\theta)\|^2, \tag{17}$$

where $P(u, v; \theta)$ is the point on the ray $(O(u, v), d(u, v))$ closest to X^{cam}, computed as

$$P = O + \big((X^{\text{cam}} - O) \cdot d\big)\, d. \tag{18}$$

The objective is the squared point-to-ray perpendicular distance. The minimisation is performed with the L-BFGS-B algorithm [6], starting from the observed pixel $(u_{\text{obs}}, v_{\text{obs}})$, with bounds $[0, W-1] \times [0, H-1]$. Convergence tolerance is set to `ftol`$=10^{-12}$, `gtol`$=10^{-8}$. For the Pycaso dataset (10 frames, 165 corners, 2 channels; 3300 observations), the full reprojection computation takes approximately 35 seconds on a single CPU core.

The 2D reprojection residual for an observation is the vector $(u_{\text{pred}} - u_{\text{obs}},\ v_{\text{pred}} - v_{\text{obs}})$ in pixels. The 2D RMS reported in Section 3.9 and Table 2 is

$$\text{RMS}_{\text{2D}} = \sqrt{\frac{1}{N}\sum_{k=1}^{N}\big[(u_{\text{pred},k} - u_{\text{obs},k})^2 + (v_{\text{pred},k} - v_{\text{obs},k})^2\big]}, \tag{19}$$

where $N = n_{\text{frames}} \times n_{\text{corners}} \times 2$ is the total number of observations.

This numerical reprojection is used as a post-optimisation diagnostic: the bundle adjustment itself minimises the ray-space residual (:func:‘point_to_ray_residuals_cmo_se3‘ in the StereoComplex codebase), which is a forward evaluation and therefore faster and smoother. After convergence, the 2D reprojection error is computed from the optimised parameters to obtain the pixel-equivalent metric reported in the paper.

Author Contributions

J.-F. Witz conceived the study, developed the StereoComplex framework, implemented all methods, conducted the experiments, and wrote the manuscript.

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During the preparation of this work the author used AI assistants (Claude, Anthropic; ChatGPT, OpenAI) to help develop and debug the Python analysis and figure-generation code, and to edit the language of the manuscript. No part of the scientific content was generated autonomously: all code, numerical results, figures, and physical interpretations were checked against the data and are the sole responsibility of the author.

Code and Data Availability

- StereoComplex: `https://github.com/jeffwitz/StereoComplex` (Code: GPL-2.0-or-later. Documentation: CC BY-SA 4.0.)
- Pycaso dataset: `https://github.com/LaboratoireMecaniqueLille/Pycaso`
- All generated artefacts: `docs/assets/pycaso_real_data/`

- Intermediate state (restart without raw images):
  `docs/assets/pycaso_real_data/intermediate_state.npz`

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